

- GENERAL NOTES:
1.

FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS
G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.
2.

PIPING DEADLEGS AND POCKETS SHALL BE ELIMINATED.
- NOTES:
1.

TAGS PREFIXED WITH GW02-WP01A.
2.

TAGS PREFIXED WITH GW02-WP01B.
3.

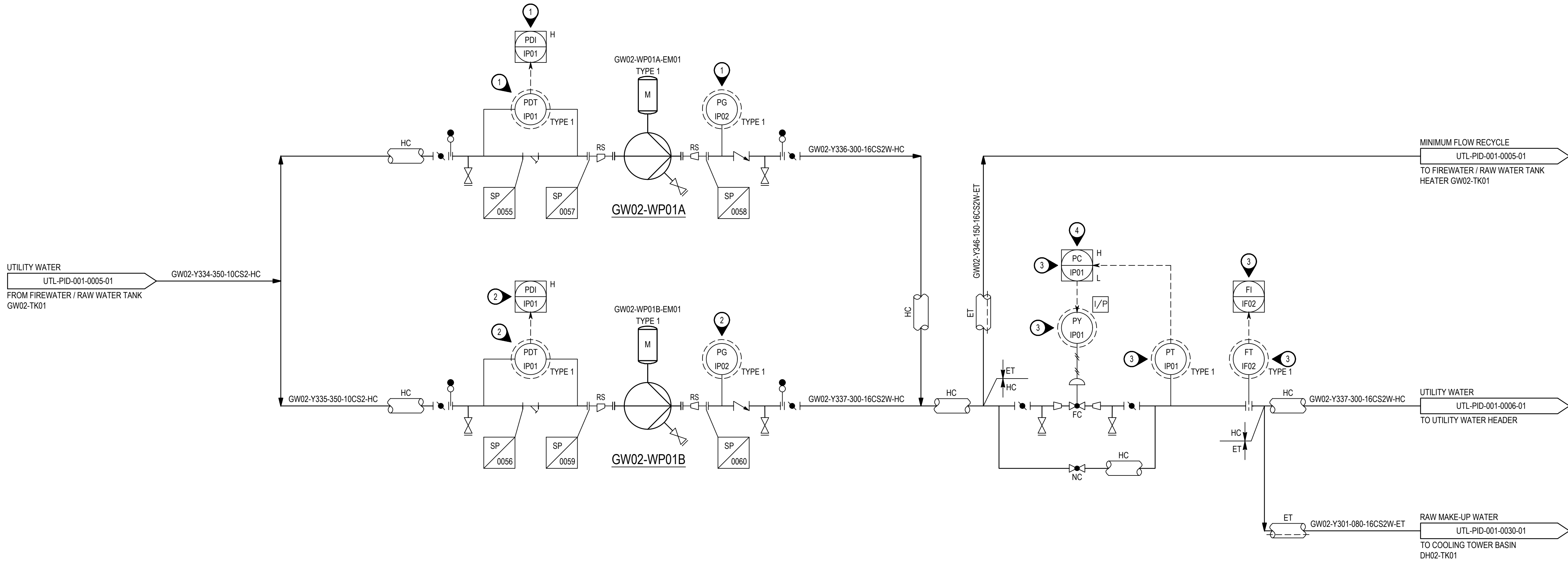
TAGS PREFIXED WITH GW02-WP01.
4.

AUTOMATIC CHANGEOVER BETWEEN DUTY / STANDBY PUMP ON DUTY PUMP FAILURE TO MAINTAIN ADEQUATE PRESSURE IN THE UTILITY WATER HEADER.
5.

2 X 100% PUMPS. DATA SHOWN IS FOR A SINGLE PUMP.

- HOLDS:
1.

DRIVER RATED POWER TO BE CONFIRMED DURING EPC.



GW02-WP01A/B
RAW / UTILITY WATER PUMPS
DESIGN CAPACITY: 668.2 m³/hr
RATED DIFFERENTIAL HEAD: 62.55 m
DRIVER RATED POWER: 250 kW (HOLD 1)
IMPELLER MATERIALS: 12% Cr
CASING MATERIALS: CS
INSULATION: ET
TRIM: Y344-16CS2W-ET (GW02-WP01 A)
Y345-16CS2W-ET (GW02-WP01 B)

THIS P&ID HAS BEEN ISSUED DIRECTLY IFH, AS IT IS SPLIT FROM G3ZR-UTL-225-PID-001-0005-01.

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

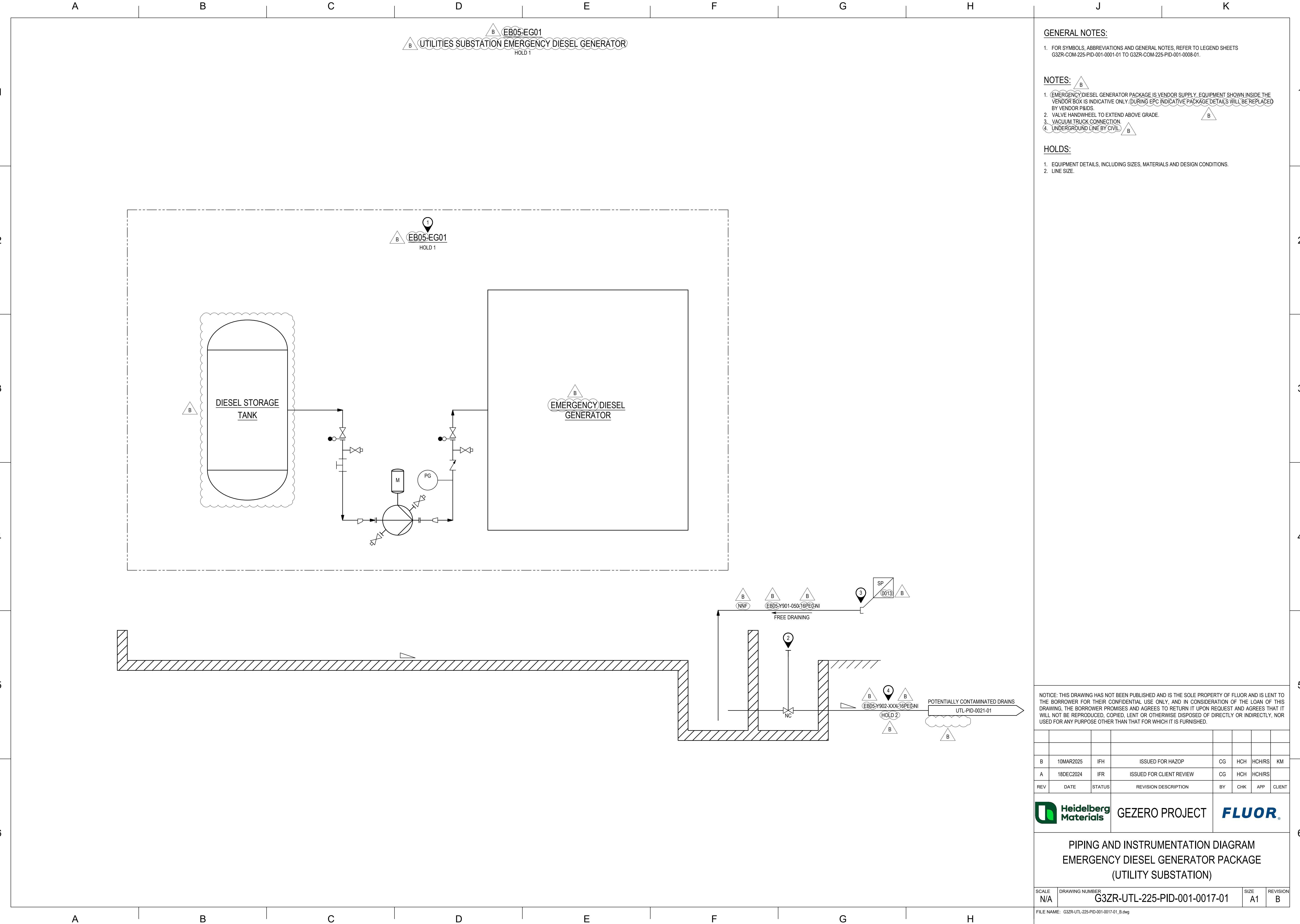
A	11MAR2025	IFH	ISSUED FOR HAZOP	AG	HCH	HCH/RS	KM
REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT



PIPING AND INSTRUMENTATION DIAGRAM
RAW / UTILITY WATER PUMPS

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-UTL-225-PID-001-0013-01	A1	A

FILE NAME: G3ZR-UTL-225-PID-001-0013-01_B.dwg



GENERAL NOTES:

1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

1. EMERGENCY DIESEL GENERATOR PACKAGE IS VENDOR SUPPLY. EQUIPMENT SHOWN INSIDE THE VENDOR BOX IS INDICATIVE ONLY. DURING EPC INDICATIVE PACKAGE DETAILS WILL BE REPLACED BY VENDOR PAIDS.
2. VALVE HANDWHEEL TO EXTEND ABOVE GRADE.
3. VACUUM TRUCK CONNECTION.
4. UNDERGROUND LINE BY CIVIL.

HOLDS:

1. EQUIPMENT DETAILS, INCLUDING SIZES, MATERIALS AND DESIGN CONDITIONS.
2. LINE SIZE.

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REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT
B	10MAR2025	IFH	ISSUED FOR HAZOP	CG	HCH	HCH/RS	KM
A	18DEC2024	IFR	ISSUED FOR CLIENT REVIEW	CG	HCH	HCH/RS	



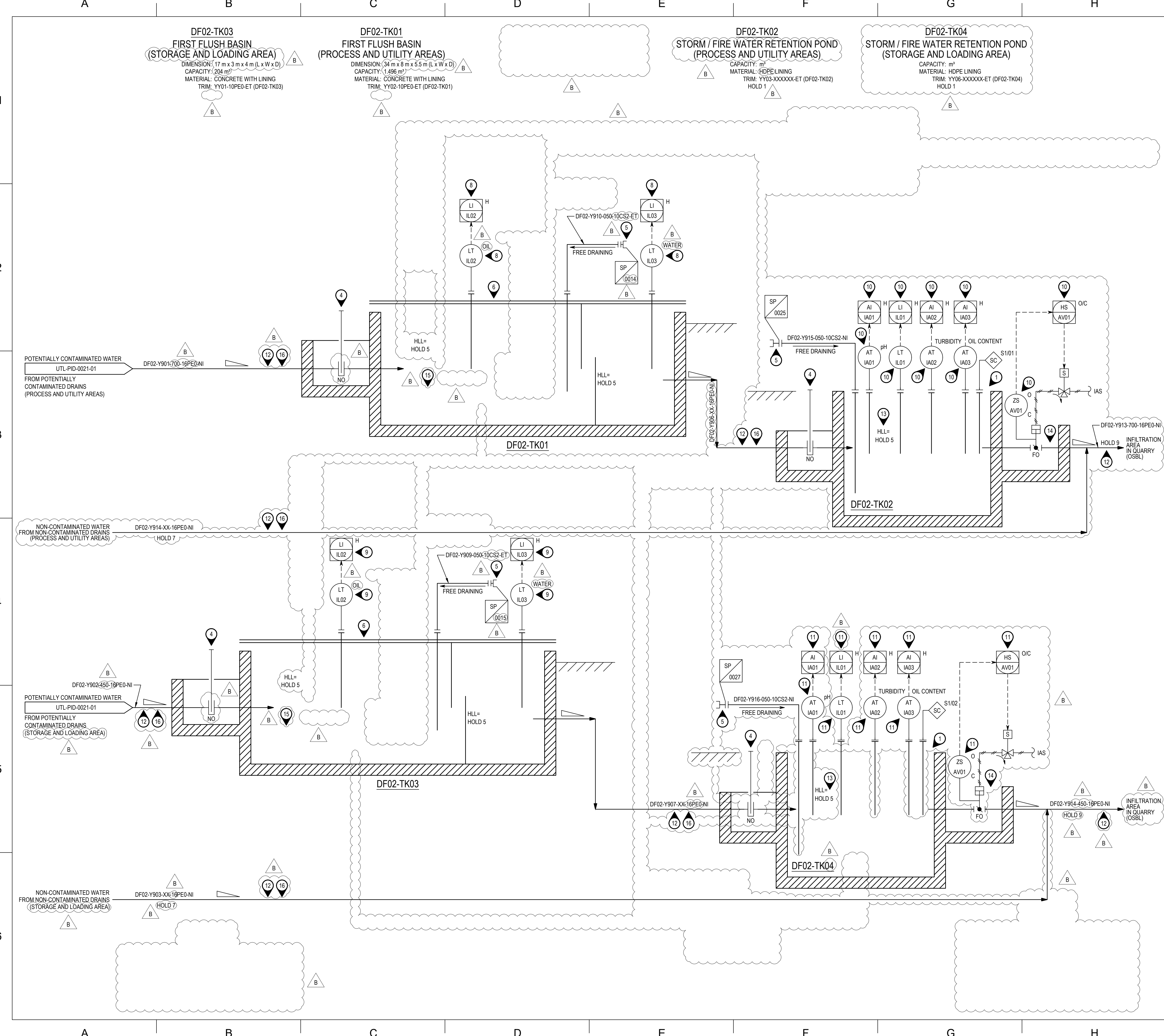
GEZERO PROJECT



PIPING AND INSTRUMENTATION DIAGRAM
EMERGENCY DIESEL GENERATOR PACKAGE
(UTILITY SUBSTATION)

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-UTL-225-PID-001-0017-01	A1	B

FILE NAME: G3ZR-UTL-225-PID-001-0017-01_B.dwg



- GENERAL NOTES:**
1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.
- NOTES:**
1. STORM / FIRE WATER POND TO BE TESTED PRIOR TO DISCHARGE. IF CLEAN, WATER CAN BE DISCHARGED TO THE FACILITY OUTFALL. IF CONTAMINATED, WATER TO BE REMOVED BY VACUUM TRUCK.
2. DELETED.
3. DELETED.
4. VALVE HANDWHEEL TO EXTEND ABOVE GRADE.
5. VACUUM TRUCK CONNECTION.
6. SUMPS TO BE PROVIDED WITH COVER TO MINIMISE DUST INGRESS.
7. DELETED.
8. TAGS PREFIXED WITH DF02-TK01.
9. TAGS PREFIXED WITH DF02-TK03.
10. TAGS PREFIXED WITH DF02-TK02.
11. TAGS PREFIXED WITH DF02-TK04.
12. UNDERGROUND LINE BY CIVIL.
13. HIGH LEVEL TO BE SELECTED TO ALLOW MAXIMUM SPACE FOR FIRE WATER VOLUME.
14. ON / OFF VALVE TO BE ACTIVATED BY THE OPERATOR (FROM THE CONTROL ROOM) WITH A TIMER TO ALARM IF VALVE HAS BEEN OPENED FOR AN EXTENDED PERIOD.
15. FIRST FLUSH BASIN TO BE DESIGNED WITH OIL OVERFLOW COMPARTMENT. OIL TO BE REMOVED BY HOSES.
16. MANHOLES TO BE DESIGNED TO HAVE SILT REMOVAL TO COLLECT SOLID / DUST PARTICULATES. OPERATING PROCEDURES TO ALLOW FOR REGULAR INSPECTION.
- HOLDS:**
1. STORM / FIRE RETENTION CAPACITY.
2. DELETED.
3. DELETED.
4. DELETED.
5. BASIN LIQUID LEVELS TO BE CONFIRMED.
6. DELETED.
7. LINE SIZES.
8. DELETED.
9. WATER QUALITY REQUIRED TO BE ALLOWED TO DISCHARGE.

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B	11MAR2025	IFH	ISSUED FOR HAZOP	CG	HCH	HCH/RS	KM
A	18DEC2024	IFR	ISSUED FOR CLIENT REVIEW	CG	HCH	HCH/RS	
REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT

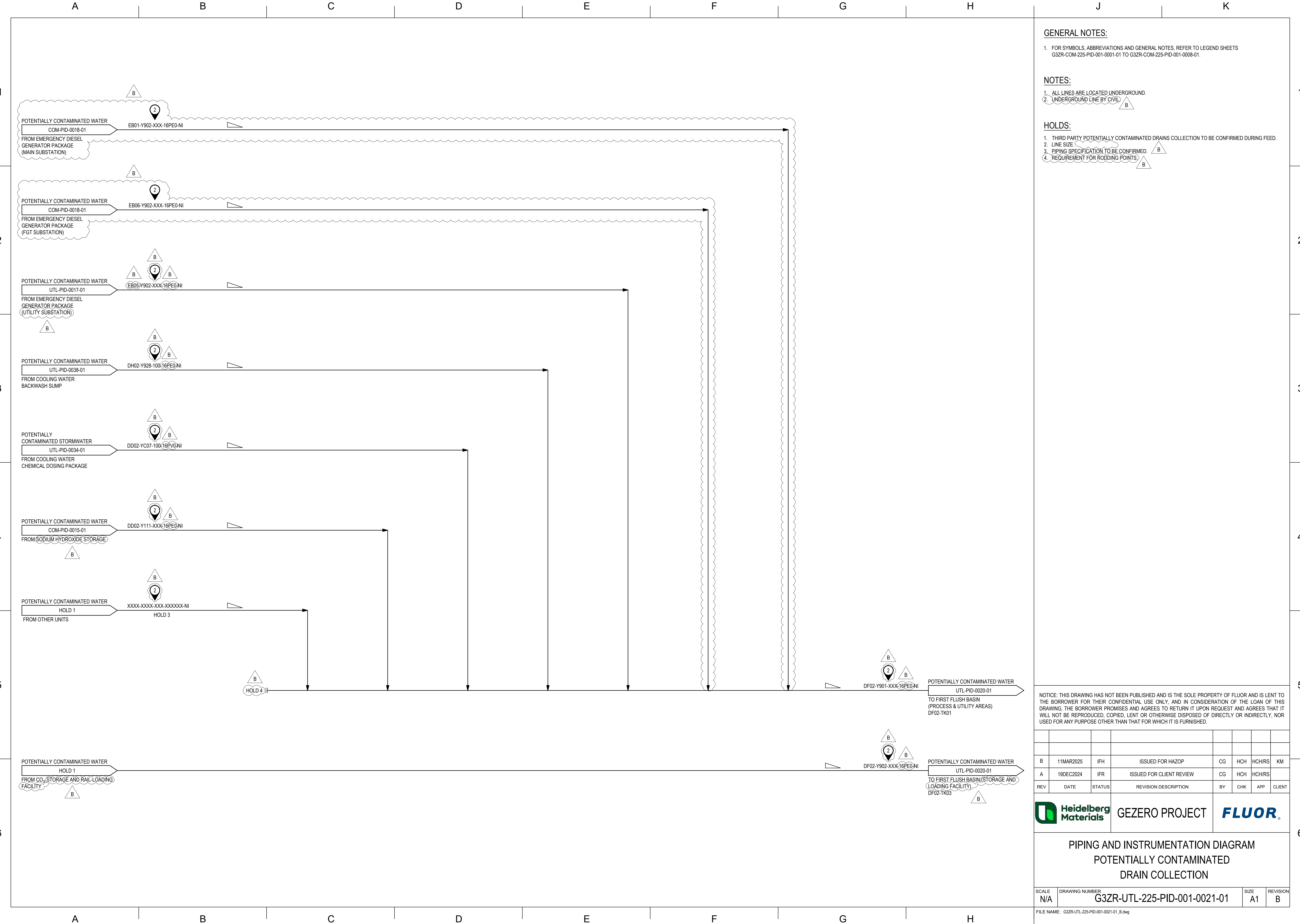
 **GEZERO PROJECT**



PIPING AND INSTRUMENTATION DIAGRAM
FIRST FLUSH BASINS AND
STORM / FIRE WATER RETENTION PONDS

SCALE N/A	DRAWING NUMBER G3ZR-UTL-225-PID-001-0020-01	SIZE A1	REVISION B
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FILE NAME: G3ZR-UTL-225-PID-001-0020-01_B.dwg



GENERAL NOTES:

1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS
G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

1. ALL LINES ARE LOCATED UNDERGROUND.
2. UNDERGROUND LINE BY CIVIL.

HOLDS:

1. THIRD PARTY POTENTIALLY CONTAMINATED DRAINS COLLECTION TO BE CONFIRMED DURING FEED.
2. LINE SIZE.
3. PIPING SPECIFICATION TO BE CONFIRMED.
4. REQUIREMENT FOR RODDING POINTS.

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A	19DEC2024	IFR	ISSUED FOR CLIENT REVIEW	CG	HCH	HCH/RS	
REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT

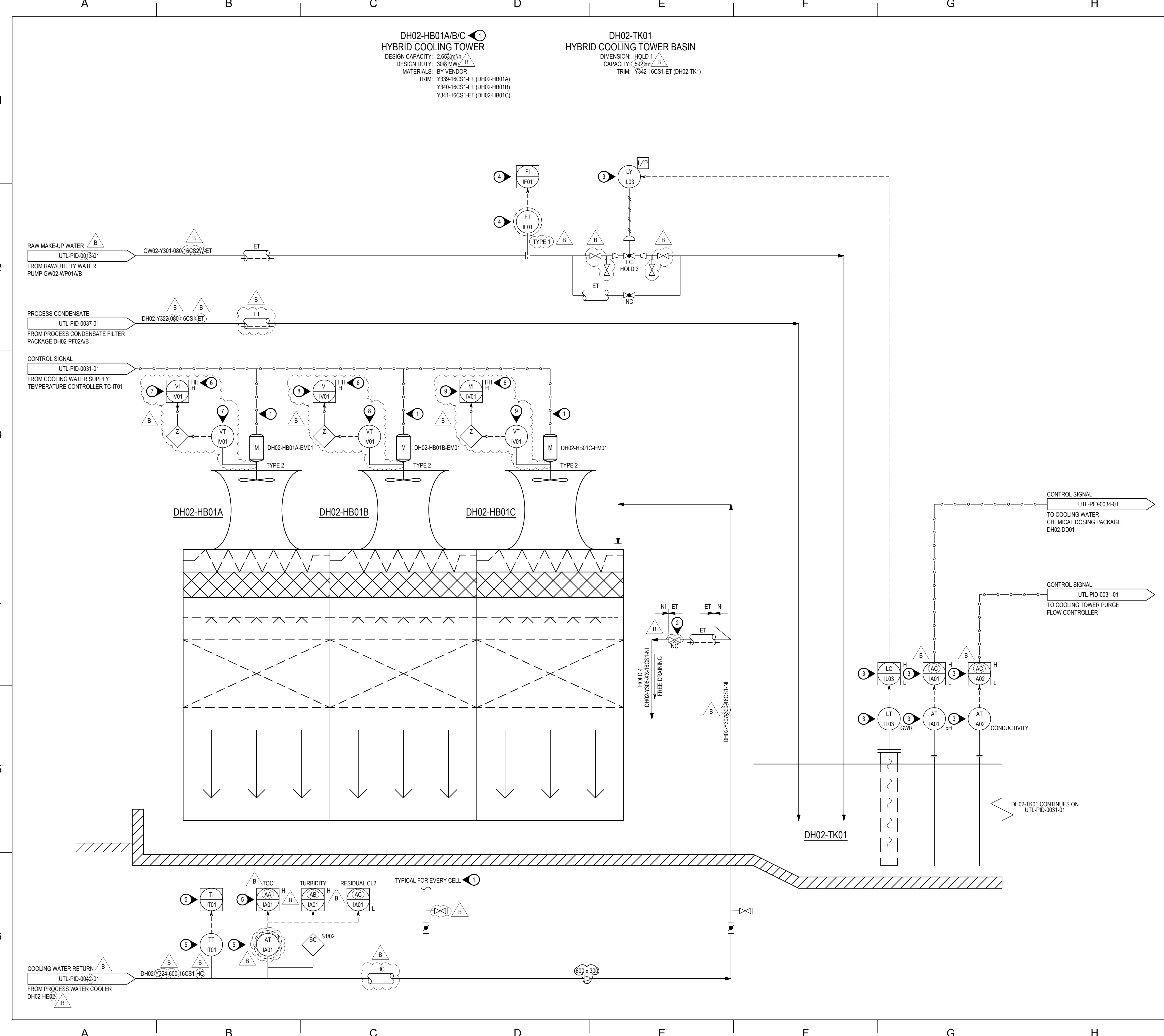


GEZERO PROJECT



PIPING AND INSTRUMENTATION DIAGRAM
POTENTIALLY CONTAMINATED
DRAIN COLLECTION

SCALE N/A	DRAWING NUMBER G3ZR-UTL-225-PID-001-0021-01	SIZE A1	REVISION B
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- GENERAL NOTES:**
- FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.
 - COOLING WATER PIPING DEADLEGS AND POCKETS SHALL BE ELIMINATED.
- NOTES:**
- 3 x 50% HYBRID COOLING TOWER. ONLY 1 RISER IS REPRESENTED IN THIS DRAWING.
 - DRAIN FROM RISER INTO COOLING TOWER BASIN. LOCATED ABOUT 0.3m ABOVE BASIN WALL AND ACCESSIBLE TO THE OPERATOR.
 - TAGS PREFIXED DH02-TK01.
 - TAGS PREFIXED GW02-WP01.
 - TAGS PREFIXED DH02-HE03.
 - MOTOR VIBRATION SWITCH TRIPS RELEVANT FAN ON HIGH-HIGH VIBRATION.
 - TAG NUMBER PREFIXED DH02-HB01A.
 - TAG NUMBER PREFIXED DH02-HB01B.
 - TAG NUMBER PREFIXED DH02-HB01C.

- HOLDS:**
- BASIN DIMENSIONS TO BE CONFIRMED.
 - INSTRUMENT SIZES INCLUDING CONTROL VALVES.
 - DELETED.
 - LINE SIZE TO BE CONFIRMED BY COOLING TOWER VENDOR. REQUIREMENT FOR ELECTRICAL TRACING TO BE CONFIRMED.

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

B	10MAR2025	IFH	ISSUED FOR HAZOP	RH	HCH	HCH/RS	KM
A	18DEC2024	IFR	ISSUED FOR CLIENT REVIEW	AG	HCH	HCH/RS	
REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT

 **GEZERO PROJECT**



PIPING AND INSTRUMENTATION DIAGRAM
HYBRID COOLING TOWER

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-UTL-225-PID-001-0030-01	A1	B

FILE NAME: G3ZR-UTL-225-PID-001-0030-01_B.dwg

GENERAL NOTES:

1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS
G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.
2. COOLING WATER PIPING DEADLEGS AND POCKETS TO BE AVOIDED.

NOTES:

1. CONDUCTIVITY LEVEL MEASURED WILL CONTROL BLOWDOWN RATE.
2. STAND-BY PUMP TO AUTO-START ON LOW PRESSURE.
3. BLOWDOWN TO BE ROUTED TO THE ADIABATIC QUENCH & CONCENTRATOR (PART OF FLUE GAS CLEANING UNIT).
4. NON SLAM CHECK VALVE TO AVOID WATER HAMMER.
5. SURGE VOLUME IS PROVIDED TO ACCUMULATE PROCESS CONDENSATE DURING PEAK WINTER AMBIENT TEMPERATURE.
6. DH02-TK01 TO BE PROVIDED WITH COVER TO MINIMISE DUST INGRESS.
7. INSTRUMENT TAGS PREFIXED DH02-TK01.
8. INSTRUMENT TAGS PREFIXED DH02-WP01A.
9. INSTRUMENT TAGS PREFIXED DH02-WP01B.
10. INSTRUMENT TAGS PREFIXED DH02-WP01C.
11. INSTRUMENT TAGS PREFIXED DH02-PF01.
12. INSTRUMENT TAGS PREFIXED DH02-PF01.
13. LINE TO BE RUN BY THIRD PARTY SITE CONTRACTOR.

HOLDS:

1. DELETED
2. INSTRUMENT SIZES INCLUDING CONTROL VALVES.
3. DELETED
4. BASIN LIQUID LEVELS TO BE CONFIRMED.
5. METHOD OF CHEMICAL DOSING MIXING WITHIN BASIN TO BE CONFIRMED DURING EPC.
6. REQUIREMENT FOR PSV TO BE CONFIRMED.
7. BEARING VIBRATION MONITORING.
8. REQUIREMENT FOR HIGH LEVEL OVERRIDE TO PREVENT OVERFILLING OF THE COOLING TOWER BASIN IN THE EVENT OF EXCESSIVE QUANTITIES OF PROCESS CONDENSATE.
9. DRIVER RATED POWER TO BE CONFIRMED DURING EPC.
10. CHEMICAL COMPATIBILITY TO BE CONFIRMED DURING EPC.

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B	10MAR2025	IFH	ISSUED FOR HAZOP	RH	HCH	HCH/RS	KM
A	19DEC2024	IFR	ISSUED FOR CLIENT REVIEW	AG	HCH	HCH/RS	
REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT

	GEZERO PROJECT	
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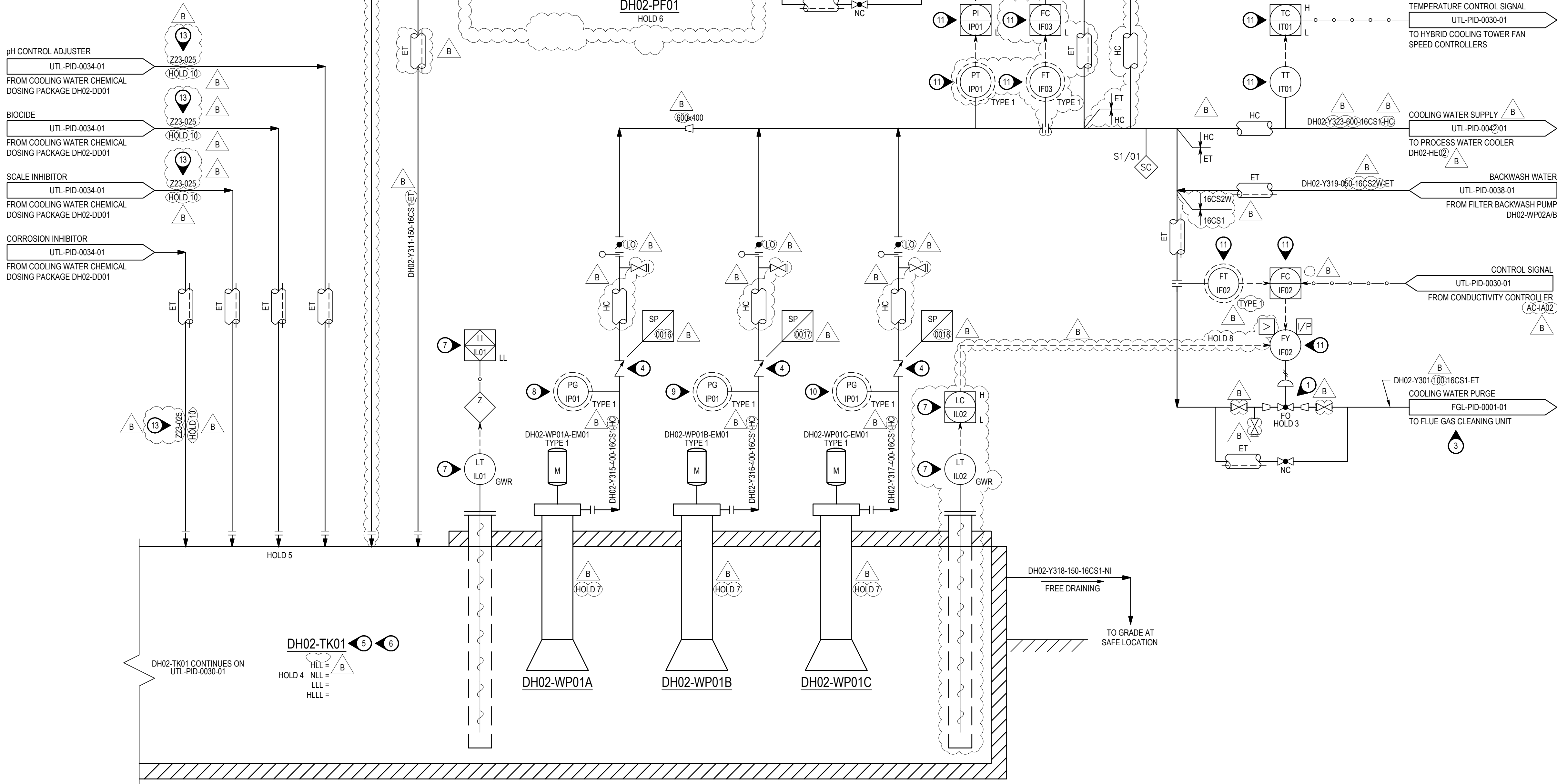
PIPING AND INSTRUMENTATION DIAGRAM
COOLING WATER CIRCULATION PUMPS
AND SIDE STREAM FILTER

SCALE N/A	DRAWING NUMBER G3ZR-UTL-225-PID-001-0031-01	SIZE A1	REVISION B
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FILE NAME: G3ZR-UTL-225-PID-001-0031-01_B.dwg

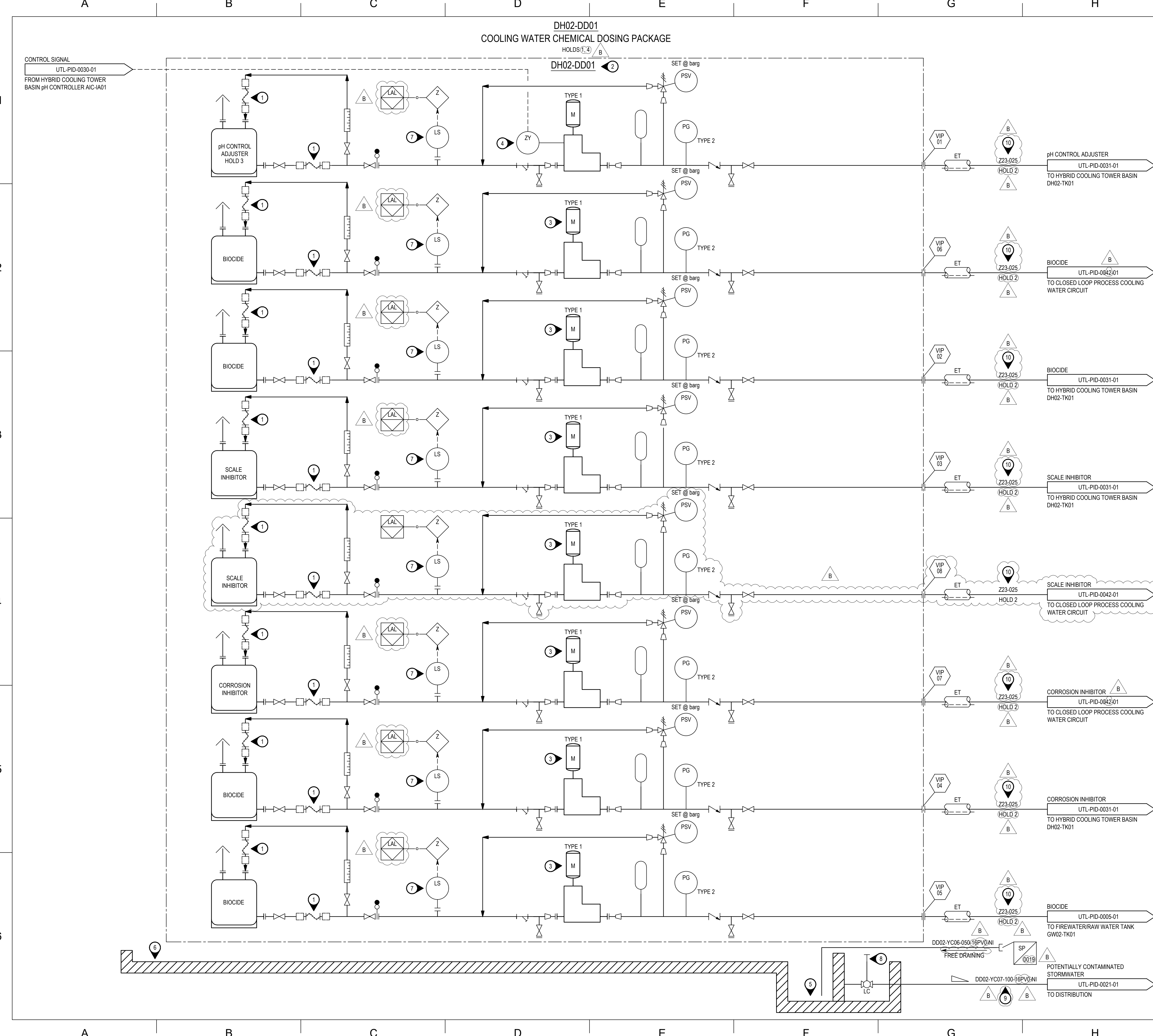
DH02-PF01
COOLING WATER SIDESTREAM FILTER

DESIGN FLOW RATE: 145.6m³/h
DESIGN PRESSURE: 15 barg
DESIGN TEMPERATURE: -25 °C / 70 °C
BODY MATERIALS: CS
FILTER MATERIALS: 316 SS
INSULATION: ET
TRIM: Y323-16CS1-ET (DH02-PF01)



DH02-WP01A/B/C
COOLING WATER CIRCULATION PUMP
(3 x 50% PUMP)

DESIGN CAPACITY: 1,544m³/h
RATED DIFFERENTIAL HEAD: 34m
DRIVER RATED POWER: 225 kW (HOLD 9)
CASING MATERIALS: CS
IMPELLER MATERIALS: 12% Cr
INSULATION: HC
TRIM: Y346-16CS1-ET (DH02-WP01A)
Y347-16CS1-ET (DH02-WP01B)
Y348-16CS1-ET (DH02-WP01C)



GENERAL NOTES:

1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

1. QUICK DISCONNECT HOSE WITH DRY BREAK COUPLING.
2. CHEMICAL DOSING PACKAGE IS VENDOR SUPPLY. THE EQUIPMENT INSIDE THE PACKAGE IS INDICATIVE ONLY.
3. MANUAL STROKE CONTROLLER.
4. ELECTRONIC STROKE CONTROLLER.
5. WATER TO BE TESTED PRIOR TO DRAINING. IF UNSUITABLE FOR DRAINING TO POTENTIALLY CONTAMINATED DRAIN SYSTEM, IT WILL BE COLLECTED BY VACUUM TRUCK FOR SAFE DISPOSAL.
6. BUND TO BE SEGREGATED SUCH THAT EACH CHEMICAL IS DRAINED INTO AN INDIVIDUAL COMPARTMENT.
7. DRY RUN PROTECTION OF CHEMICAL DOSING PUMP TO BE PROVIDED BY LEVEL MEASUREMENT OF VIBRATION FORK.
8. VALVE HAND WHEEL TO EXTEND ABOVE GRADE.
9. UNDERGROUND LINE BY CIVIL.
10. LINE TO BE RUN BY THIRD PARTY SITE CONTRACTOR.

HOLDS:

1. NUMBER AND TYPE OF CHEMICALS TO BE CONFIRMED DURING EPC.
2. CHEMICAL COMPATIBILITY TO BE CONFIRMED DURING EPC.
3. pH CONTROL ADJUSTER TYPE PENDING FINAL DCG PROCESS CONDENSATE COMPOSITION.
4. EPC TO CONFIRM WHETHER DUPLICATE DOSING SKIDS ARE REQUIRED FOR BIOCIDES AND CORROSION INHIBITOR OR WHETHER A SINGLE DOSING SYSTEM IS NEEDED.

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B	10MAR2025	IFH	ISSUED FOR HAZOP	RH	HCH	HCH/RS	KM
A	19DEC2024	IFR	ISSUED FOR CLIENT REVIEW	AG	HCH	HCH/RS	
REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT

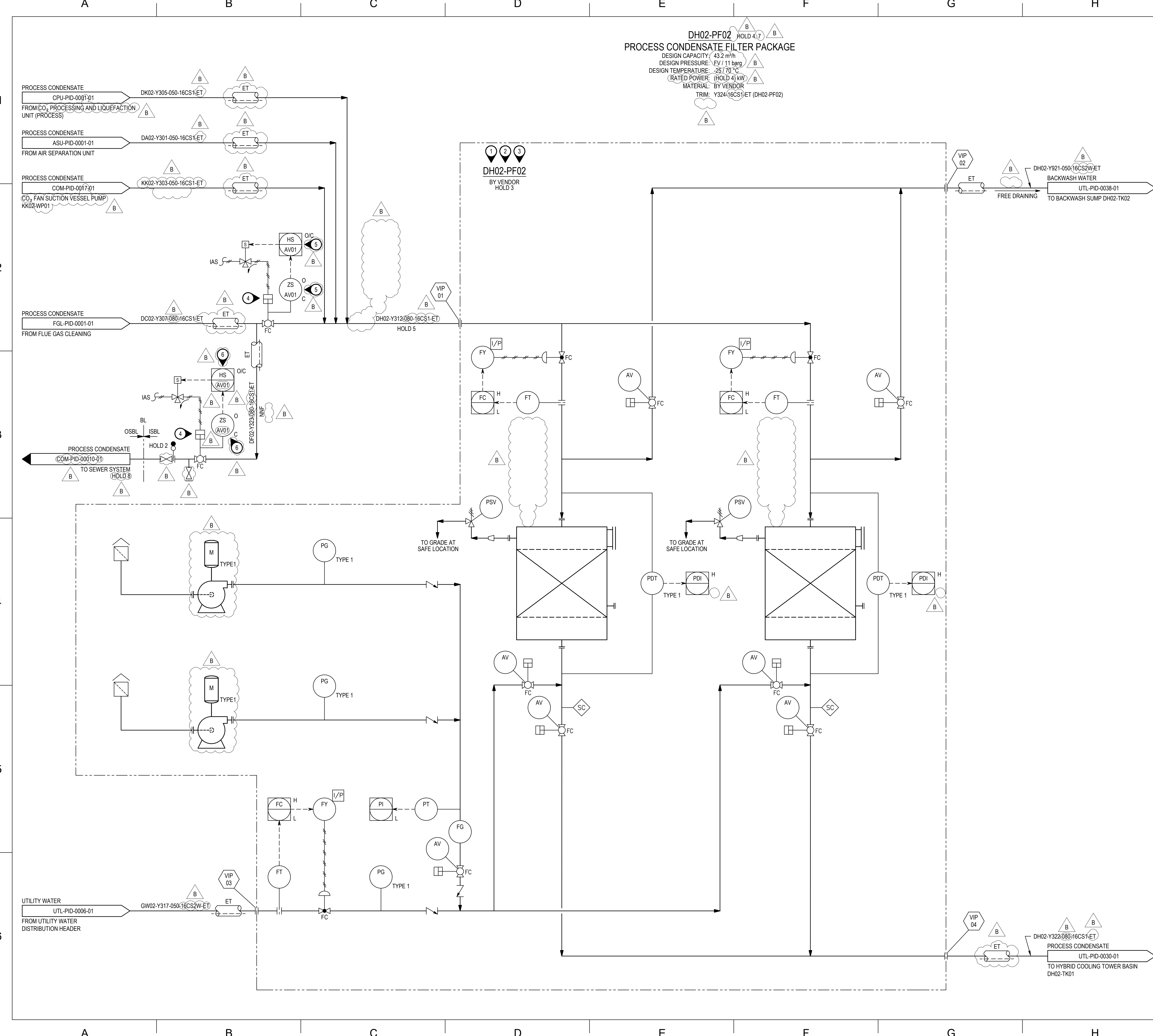
**GEZERO PROJECT**



PIPING AND INSTRUMENTATION DIAGRAM
COOLING WATER CHEMICAL
DOSING PACKAGE

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-UTL-225-PID-001-0034-01	A1	B

FILE NAME: G3ZR-UTL-225-PID-001-0034-01_B.dwg



DH02-PF02
PROCESS CONDENSATE FILTER PACKAGE
DESIGN CAPACITY: 43.2 m³/h
DESIGN PRESSURE: 11 barg
DESIGN TEMPERATURE: 25/70 °C
(RATED POWER: (HOLD 4) kW)
MATERIAL: BY VENDOR
TRIM: Y324-16CS1-ET (DH02-PF02)

- GENERAL NOTES:**
- FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.
 - PIPING DEADLEGS AND POCKETS SHALL BE ELIMINATED.

- NOTES:**
- PROCESS CONDENSATE FILTER PACKAGE IS VENDOR SUPPLY. THE EQUIPMENT INSIDE THE PACKAGE IS INDICATIVE ONLY.
 - PACKAGE ISOLATION BY VENDOR.
 - PACKAGE TO CONTAIN FACILITIES FOR DRAINING UTILITY WATER TO PREVENT STAGNANT CONDITIONS.
 - ROUTING VALVES USED TO DIVERT DCC PROCESS CONDENSATE TO THE SEWER SYSTEM IN THE EVENT OF A PROCESS UPSET CONDITION.
 - TAGS PREFIXED DC02-PI01.
 - TAGS PREFIXED DF02-PI01.

- HOLDS:**
- DELETED.
 - TIE-IN BATTERY LIMIT ISOLATION.
 - VENDOR PAID NUMBER.
 - EQUIPMENT DETAILS INCLUDING SIZES, MATERIALS, DESIGN CONDITIONS AND RELIEF REQUIREMENTS.
 - DELETED.
 - DELETED.
 - FILTER TYPE TO BE CONFIRMED.

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REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT

 **GEZERO PROJECT**



PIPING AND INSTRUMENTATION DIAGRAM
PROCESS CONDENSATE FILTER PACKAGE

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-UTL-225-PID-001-0037-01	A1	B

FILE NAME: G3ZR-UTL-225-PID-001-0037-01_B.dwg

DH02-TK02 HOLD 3
FILTER BACKWASH SUMP
DIMENSIONS: / HOLD 1
CAPACITY: / HOLD 1
TRIM: Y930-16CS2W-ET (DH02-TK02)

GENERAL NOTES:

- FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.
- PIPING DEADLEGS AND POCKETS SHALL BE ELIMINATED.

NOTES:

- CONNECTION FOR EMPTYING SUMP BY VACUUM TRUCK.
- PROVIDE LOW POINT IN SUMP FOR SETTLEMENT OF SOLIDS.
- SUMP TO BE SLOPED TOWARDS PUMP SUCTION.
- THE PUMP WILL OPERATE AS FOLLOWS:
 - AT HIGH HIGH LEVEL, START STAND-BY AND STOP DUTY PUMP.
 - AT HIGH LEVEL, START PUMP.
 - AT LOW LEVEL, STOP PUMP.
 - AT LOW LOW LEVEL, TRIP PUMP MOTOR FOR DRY RUN PROTECTION.
- 2 x 100% SELF PRIMING MOTOR DRIVEN HORIZONTAL CENTRIFUGAL PUMP.
- RESTRICTION ORIFICE TO BE HIGH POINT.
- DH02-TK02 TO BE PROVIDED WITH COVER TO MINIMISE DUST INGRESS.
- TAGS PREFIXED DH02-TK02.
- TAGS PREFIXED DH02-WP02A.
- TAGS PREFIXED DH02-WP02B.
- TAGS PREFIXED DH02-WP02.
- HIGH AND LOW ALARMS TO BE SET OUTSIDE THE THRESHOLD OF PUMP START / STOP.
- VALVE HAND WHEEL TO EXTEND ABOVE GRADE.
- UNDERGROUND LINE BY CIVIL.

HOLDS:

- SUMP CAPACITY AND DIMENSIONS TO BE CONFIRMED BASED ON BACKWASH DURATION AND FREQUENCY.
- DELETED.
- LIQUID LEVELS.
- REQUIREMENT FOR FACILITIES FOR HANDLING PARTICULATES IN PUMP SUCTION.
- DELETED.
- REQUIREMENT FOR FILTER BACKWASH SUMP AND PUMPS.
- DELETED.
- PUMP TYPE AND MATERIALS OF CONSTRUCTION TO BE CONFIRMED.

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REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT



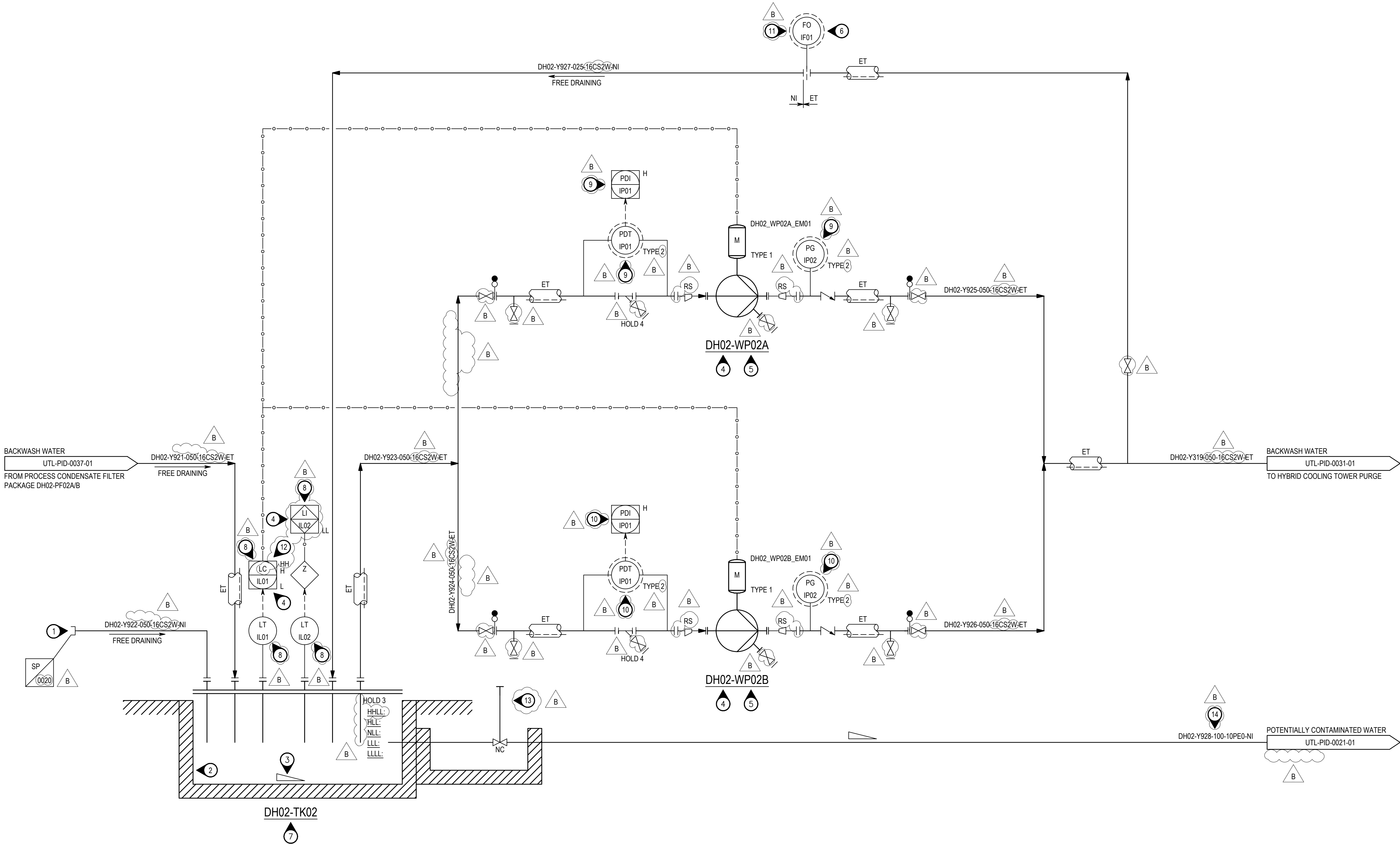
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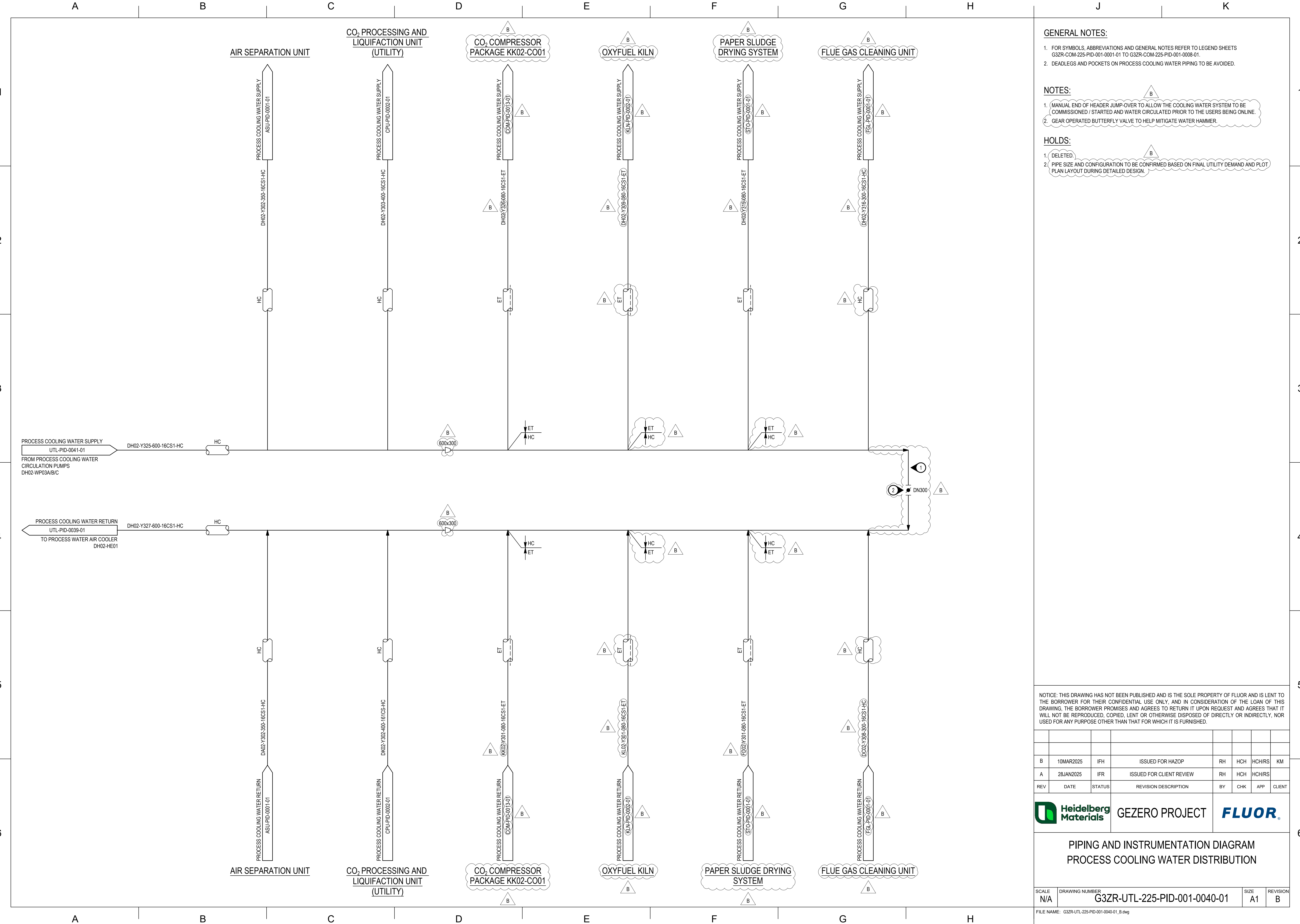
PIPING AND INSTRUMENTATION DIAGRAM
BACKWASH SUMP AND PUMPS

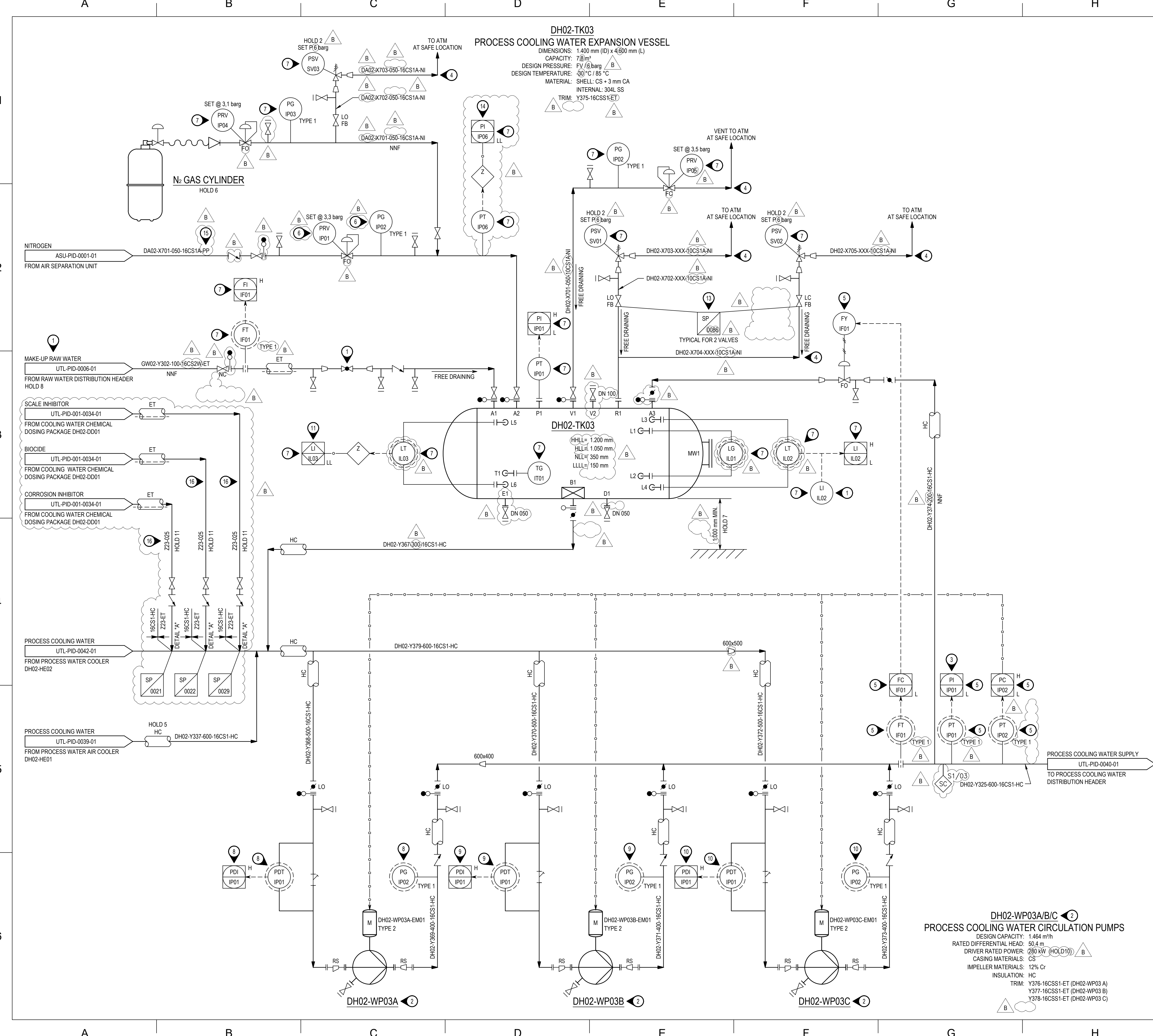
SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-UTL-225-PID-001-0038-01	A1	B

FILE NAME: G3ZR-UTL-225-PID-001-0038-01_B.dwg



DH02-WP02A/B HOLDS 6, 8
FILTER BACKWASH PUMP
DESIGN CAPACITY: 6.5 m³/h
RATED DIFFERENTIAL HEAD: 38.6 m
DRIVER RATED POWER: 0.7 kW
CASING MATERIALS: CS (HOLD 8)
IMPELLER MATERIALS: 12% Cr (HOLD 8)
INSULATION: ET
TRIM: Y929-16CS2W-ET (DH02-WP02A)
Y930-16CS2W-ET (DH02-WP02B)





GENERAL NOTES:

- FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-001-01 TO G3ZR-COM-225-PID-001-0008-01.
- DEADLEGS AND POCKETS ON PROCESS COOLING WATER PIPING TO BE AVOIDED.

NOTES:

- THE LEVEL INDICATOR SHALL BE VISIBLE FROM MAKEUP (FILLING) GLOBE VALVE.
- 3 x 50% MOTOR DRIVEN CENTRIFUGAL PUMPS. DATA SHOWN IS FOR A SINGLE PUMP.
- STAND-BY PUMP TO AUTO-START ON LOW PRESSURE.
- WEEP HOLE TO BE PROVIDED AT LOW POINT.
- TAG NUMBER PREFIXED DH02-WP03.
- TAG NUMBER PREFIXED DA02-PI01.
- TAG NUMBER PREFIXED DH02-TK03.
- TAG NUMBER PREFIXED DH02-WP03A.
- TAG NUMBER PREFIXED DH02-WP03B.
- TAG NUMBER PREFIXED DH02-WP03C.
- DRY RUN PROTECTION OF THE PUMPS TO BE PROVIDED BY LEVEL TRANSMITTER LT-IL03.
- DELETED.
- KEY OPERATED INTERLOCK SYSTEM.
- LOW LOW NITROGEN PRESSURE TO STOP THE PUMPS AND PROTECT AGAINST A LOSS OF NPSHA.
- LINE TO BE PROVIDED WITH PERSONNEL PROTECTION TO PROTECT OPERATORS IN THE EVENT NITROGEN IS SUPPLIED FROM THE LIQUID NITROGEN STORAGE DURING THE COLDEST PERIODS OF THE YEAR.
- LINE TO BE RUN BY THIRD PARTY SITE CONTRACTOR.

HOLDS:

- DELETED.
- PSV LINE, INLET AND OUTLET SIZE.
- DELETED.
- DELETED.
- REQUIREMENT FOR LINE DURING WINTER OPERATION OF THE COOLING WATER SYSTEM TO BE CONFIRMED.
- NUMBER OF NITROGEN GAS CYLINDERS REQUIRED.
- ELEVATION ABOVE GRADE.
- HM TO CONFIRM PRESSURE OF RAW WATER TIE-IN.
- DELETED.
- DRIVER RATED POWER TO BE CONFIRMED DURING EPC.
- CHEMICAL COMPATIBILITY TO BE CONFIRMED DURING EPC.

DETAIL "A"

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

B	11MAR2025	IFH	ISSUED FOR HAZOP	RH	HCH	HCH/RS	KM
A	28JAN2025	IFR	ISSUED FOR CLIENT REVIEW	RH	HCH	HCH/RS	
REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT

GEZERO PROJECT

PIPING AND INSTRUMENTATION DIAGRAM
PROCESS COOLING WATER
EXPANSION VESSEL AND CIRCULATION PUMPS

SCALE N/A	DRAWING NUMBER G3ZR-UTL-225-PID-001-0041-01	SIZE A1	REVISION B
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FILE NAME: G3ZR-UTL-225-PID-001-0041-01_B.dwg

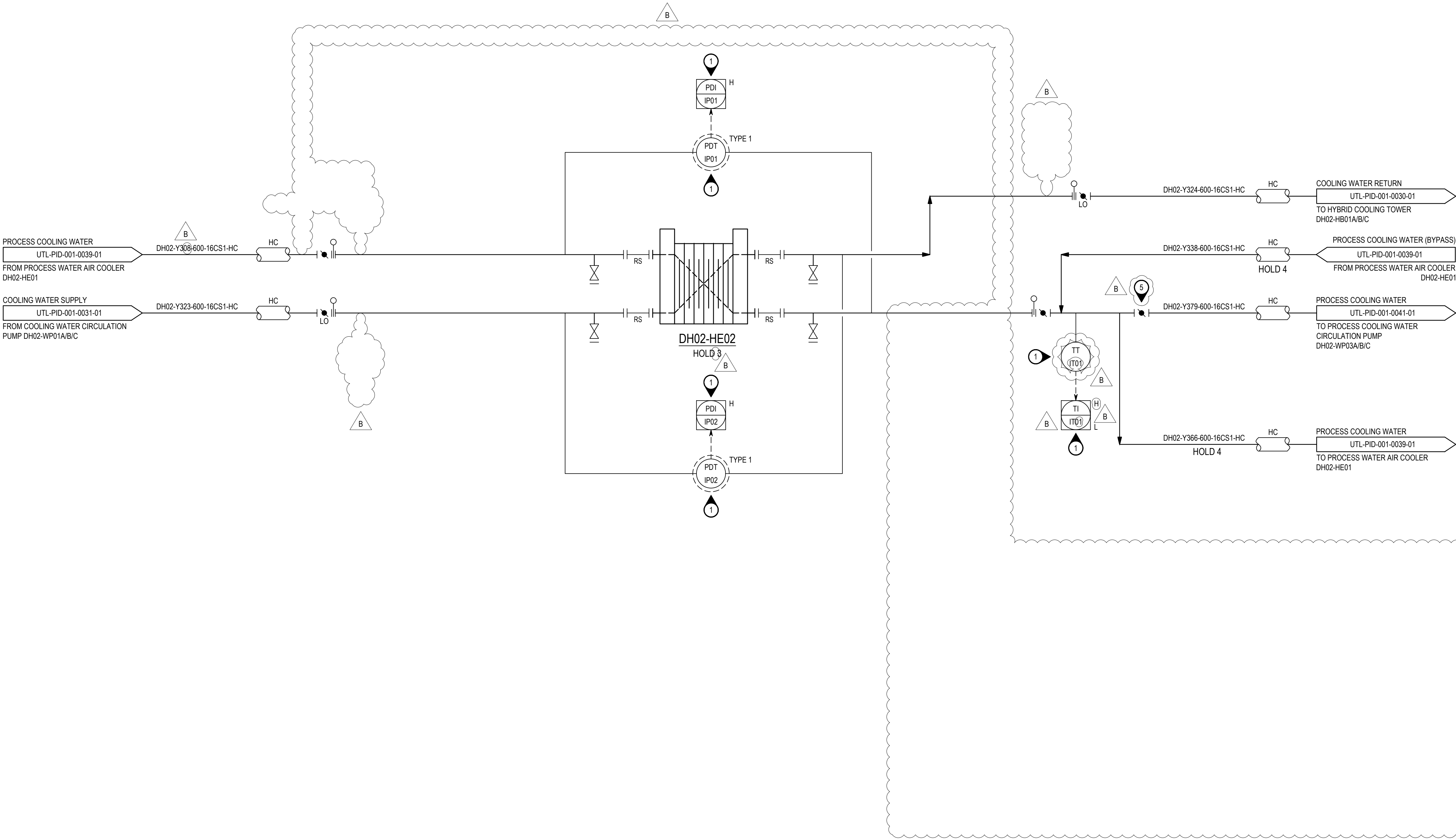
DH02-WP03A/B/C
PROCESS COOLING WATER CIRCULATION PUMPS

DESIGN CAPACITY: 1,464 m³/h
RATED DIFFERENTIAL HEAD: 50.4 m
DRIVER RATED POWER: 280 kW (HOLD)
CASING MATERIALS: CS
IMPELLER MATERIALS: 12% Cr
INSULATION: HC
TRIM: Y378-16CSS1-ET (DH02-WP03 A)
Y377-16CSS1-ET (DH02-WP03 B)
Y378-16CSS1-ET (DH02-WP03 C)

DH02-HE02
PROCESS WATER COOLER
DESIGN DUTY: 30.8 MW
DESIGN PRESSURE: 11 barg
DESIGN TEMPERATURE: -25 °C / 85 °C
MATERIAL: PLATE: DIN 1.4301 (304SS)
FRAME: CS + 3mm CA
TRIM: Y381-16CS1-ET
HOLD 3

- GENERAL NOTES:
- FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.
 - DEADLEGS AND POCKETS ON PROCESS COOLING WATER PIPING TO BE AVOIDED.

- NOTES:
- TAGS PREFIXED DH02-HE02.
- HOLDS:
- DELETED.
 - DELETED.
 - NUMBER OF PLATE AND FRAME HEAT EXCHANGERS TO BE CONFIRMED.
 - REQUIREMENT FOR LINE DURING WINTER OPERATION OF THE COOLING WATER SYSTEM TO BE CONFIRMED.
 - VALVE TO BE USED FOR ROUTING DURING WINTER MODE OF OPERATION.



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REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT
B	10MAR2025	IFH	ISSUED FOR HAZOP	RH	HCH	HCH/RS	KM
A	28JAN2025	IFR	ISSUED FOR CLIENT REVIEW	RH	HCH	HCH/RS	



GEZERO PROJECT



PIPING AND INSTRUMENTATION DIAGRAM
PROCESS WATER COOLER

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-UTL-225-PID-001-0042-01	A1	B

FILE NAME: G3ZR-UTL-225-PID-001-0042-01_B.dwg

GENERAL NOTES:

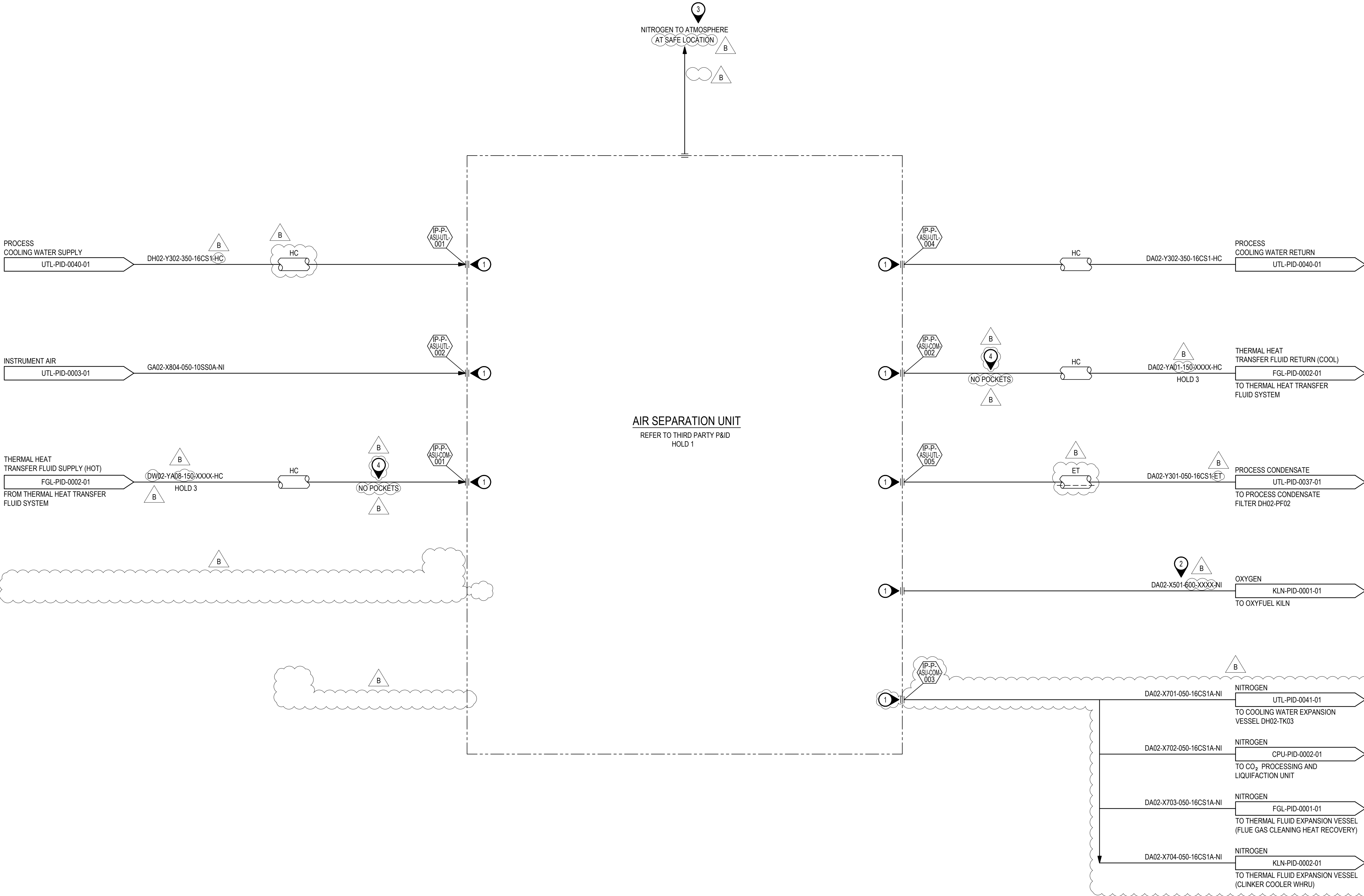
1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS
G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

1. PACKAGE ISOLATION IS WITHIN THIRD PARTY (ASU) SCOPE.
2. LINE IS WITHIN THIRD PARTY (ASU) SCOPE.
3. NITROGEN VENT TO ATMOSPHERE WITHIN THIRD PARTY (ASU) AREA.
4. ALL THERMAL FLUID LINES AND EQUIPMENT TO BE FREE DRAINING TO COLLECTION VESSELS FROM HIGH POINT (HOLD 6).

HOLDS:

1. THIRD PARTY P&ID NUMBER(S).
2. DELETED.
3. THERMAL HEAT TRANSFER FLUID PIPE SIZES AND SPECIFICATIONS BY THIRD PARTY (FLUE GAS CLEANING).
4. DELETED.
5. DELETED.
6. LOCATION OF THERMAL FLUID DRAIN VESSELS AND THEREFORE PIPING HIGH POINT LOCATIONS.
7. EPC CONTRACTOR TO CONVERT PIPING MATERIAL CLASSES PROVIDED BY THIRD PARTIES TO A CONSISTENT PROJECT PIPING CLASS INDEX AND RESOLVE ANY DISCREPANCIES.



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B	10MAR2025	IFH	ISSUED FOR HAZOP	PB	HCH	HCH/RS	KM
A	19DEC2024	IFR	ISSUED FOR CLIENT REVIEW	AG	HCH	HCH/RS	



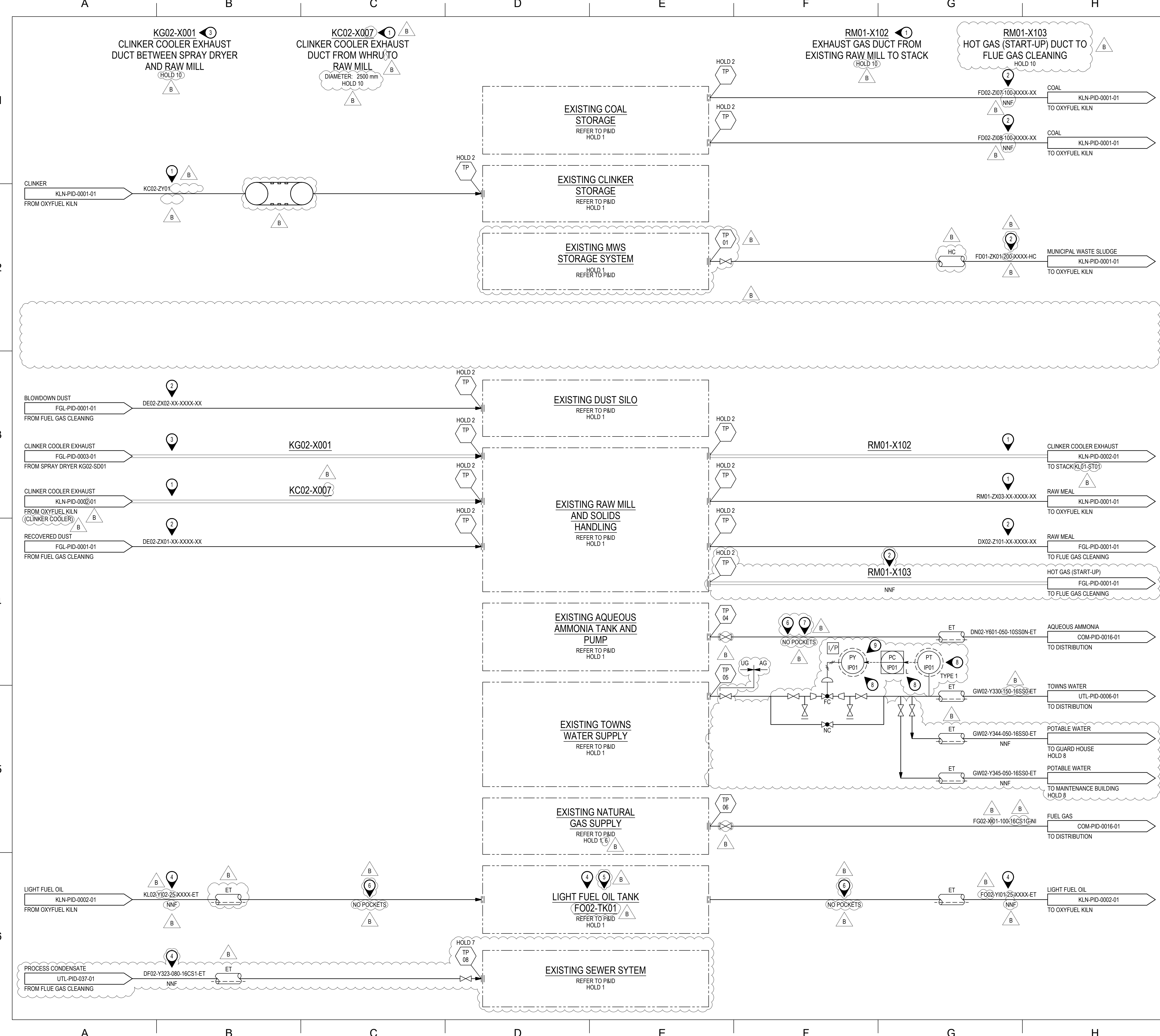
GEZERO PROJECT



PIPING AND INSTRUMENTATION DIAGRAM
AIR SEPARATION UNIT INTERFACES

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-ASU-225-PID-001-0001-01	A1	B

FILE NAME: G3ZR-ASU-225-PID-001-0001-01_B.dwg



GENERAL NOTES:

1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

1. TRANSFER LINE / DUCT IS WITHIN THIRD PARTY (OXYFUEL KILN) SCOPE. DETAILS BY THIRD PARTY.
2. TRANSFER LINE IS WITHIN HM SCOPE.
3. TRANSFER LINE / DUCT IS WITHIN THIRD PARTY (FLUE GAS CLEANING) SCOPE. DETAILS BY THIRD PARTY.
4. LIGHT FUEL OIL TANK AND TRANSFER LINE ARE WITHIN HM SCOPE.
5. LIGHT OIL TANK EXISTING TAG NUMBER IS 290-70.
6. LINE TO HAVE NO POCKETS TO ENABLE DRAINING TO EITHER SOURCE OR DESTINATION.
7. TOXIC LINE. MINIMISE THE USE OF FLANGES.
8. TAGS PREFIXED WITH GW02-PI01.
9. PRESSURE REGULATION REQUIRED BY LOCAL WATER AUTHORITY.

HOLDS:

1. EXISTING SYSTEM P&ID NUMBER(S).
2. THIRD-PARTY TIE-IN NUMBERS.
3. ALL TIE-IN POINT ISOLATION REQUIREMENTS.
4. (DELETED).
5. LINE SIZES, SPECIFICATION AND INSULATION.
6. REQUIREMENT FOR NATURAL GAS CONNECTION TO EXISTING KILN.
7. ACCEPTANCE OF THE DCC CONDENSATE BY THE LOCAL WATER COMPANY DURING START-UP/EMERGENCIES TO BE CONFIRMED DURING EPC.
8. BUILDING NUMBER TO BE PROVIDED DURING EPC.
9. REQUIREMENT FOR ADDITIONAL HEAT TRANSFER WATER LINE FROM MWS DRYING WATER TO / FROM MAINTENANCE BUILDING.
10. DUCT EQUIPMENT DETAILS.

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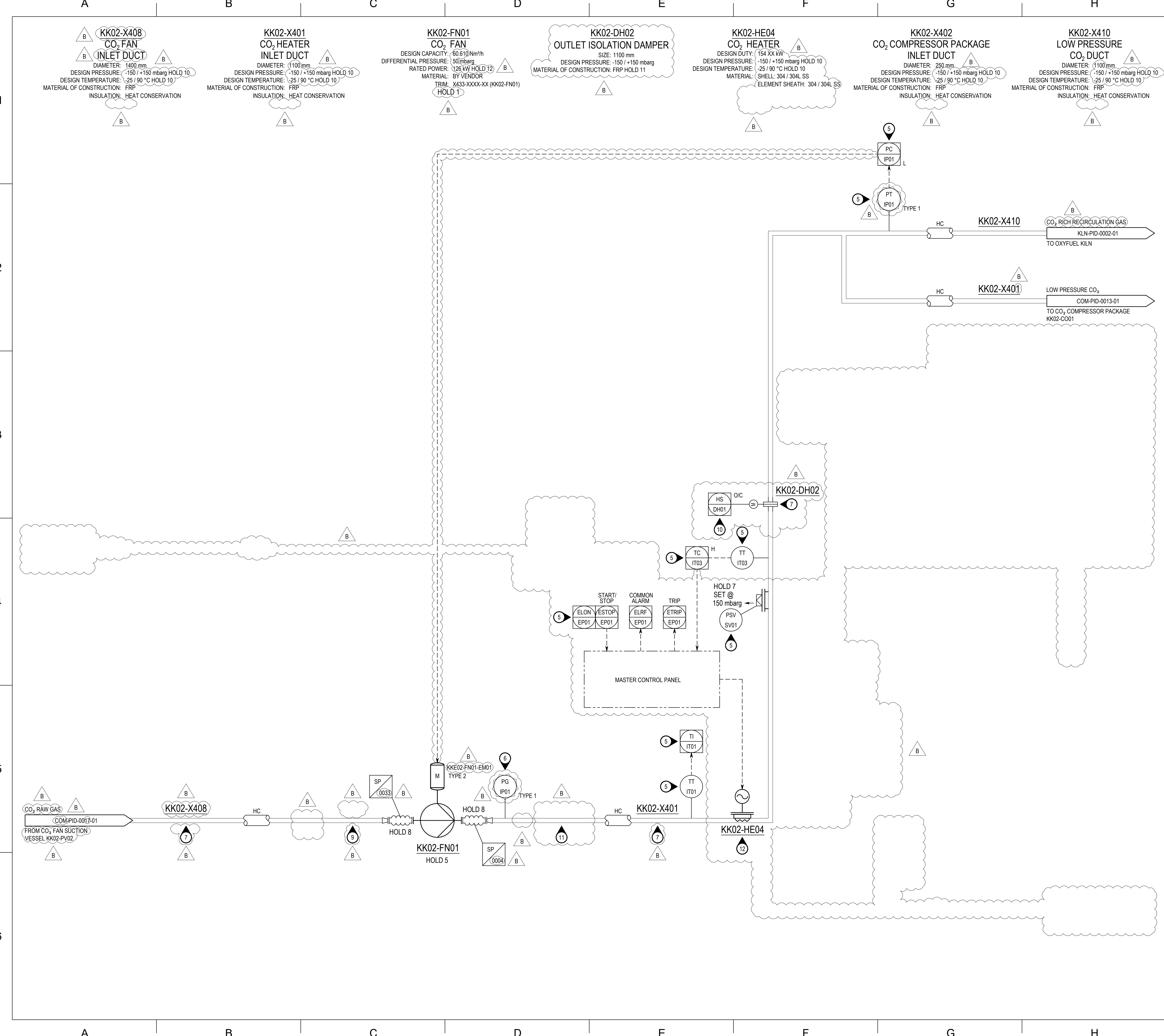
B	12MAR2025	IFH	ISSUED FOR HAZOP	PB	HCH	HCH/RS	KM		
A	19DEC2024	IFR	ISSUED FOR CLIENT REVIEW	PB	HCH	HCH/RS			
REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT		

GEZERO PROJECT

PIPING AND INSTRUMENTATION DIAGRAM
TIE-INS

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-COM-225-PID-001-0010-01	A1	B

FILE NAME: G3ZR-COM-225-PID-001-0010-01_B.dwg



GENERAL NOTES:

1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS
G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

1. (DELETED)
2. (DELETED)
3. 10mm WEEP HOLE TO BE PROVIDED AT LOWEST POINT OF PSV OUTLET / VENT LINE. LINES TO BE SLOPED TO MINIMISE RISK OF WATER COLLECTION AND SUBSEQUENT CARBONIC ACID FORMATION.
4. (DELETED)
5. TAGS PREFIXED WITH KK02_HE04.
6. TAGS PREFIXED WITH KK02_FN01.
7. DISTANCE BETWEEN THE DAMPER UPSTREAM OF THE CO₂ FAN SUCTION VESSEL AND THE DAMPER DOWNSTREAM OF THE CO₂ HEATER IS TO BE MINIMISED. THIS IS REQUIRED TO MINIMISE VOLUME OF CO₂ RAW GAS VENTED IN THE EVENT OF MAINTENANCE. CO₂ FAN SUCTION VESSEL, CO₂ FAN AND CO₂ HEATER WILL BE MAINTAINED DURING THE ANNUAL CEMENT PLANT TURNAROUND.
8. KEY OPERATED INTERLOCK SYSTEM.
9. MAINTENANCE DRAIN TO BE PROVIDED AT LOW POINT.
10. TAGS PREFIXED WITH KK02_DH02.
11. CO₂ FAN DISCHARGE TO HAVE A MINIMUM OF THREE EQUIVALENT DIAMETERS STRAIGHT LENGTH TO FIRST RESTRICTION, IN ORDER TO REDUCE TURBULENCE.
12. CO₂ HEATER TO BE DUCT STAB-IN HEATER.

HOLDS:

1. EQUIPMENT DETAILS, INCLUDING SIZES, MATERIALS AND DESIGN CONDITIONS.
2. (DELETED)
3. CONTROL AND INSTRUMENTATION INTERFACE WITH BLOWER TO BE CONFIRMED.
4. (DELETED)
5. REQUIREMENT FOR LOW PRESSURE START PERMISSIVE.
6. (DELETED)
7. RELIEF VALVE SIZE AND LINE SIZE.
8. REQUIREMENT FOR EXPANSION JOINT TO BE CONFIRMED BASED ON STRESS ANALYSIS.
9. (DELETED)
10. DUCT DESIGN CONDITIONS TO BE CONFIRMED BY THIRD PARTY (GEA).
11. MATERIAL OF CONSTRUCTION TO BE CONFIRMED BY VENDOR.
12. PRELIMINARY VALUE TO BE CONFIRMED BY THE VENDOR.

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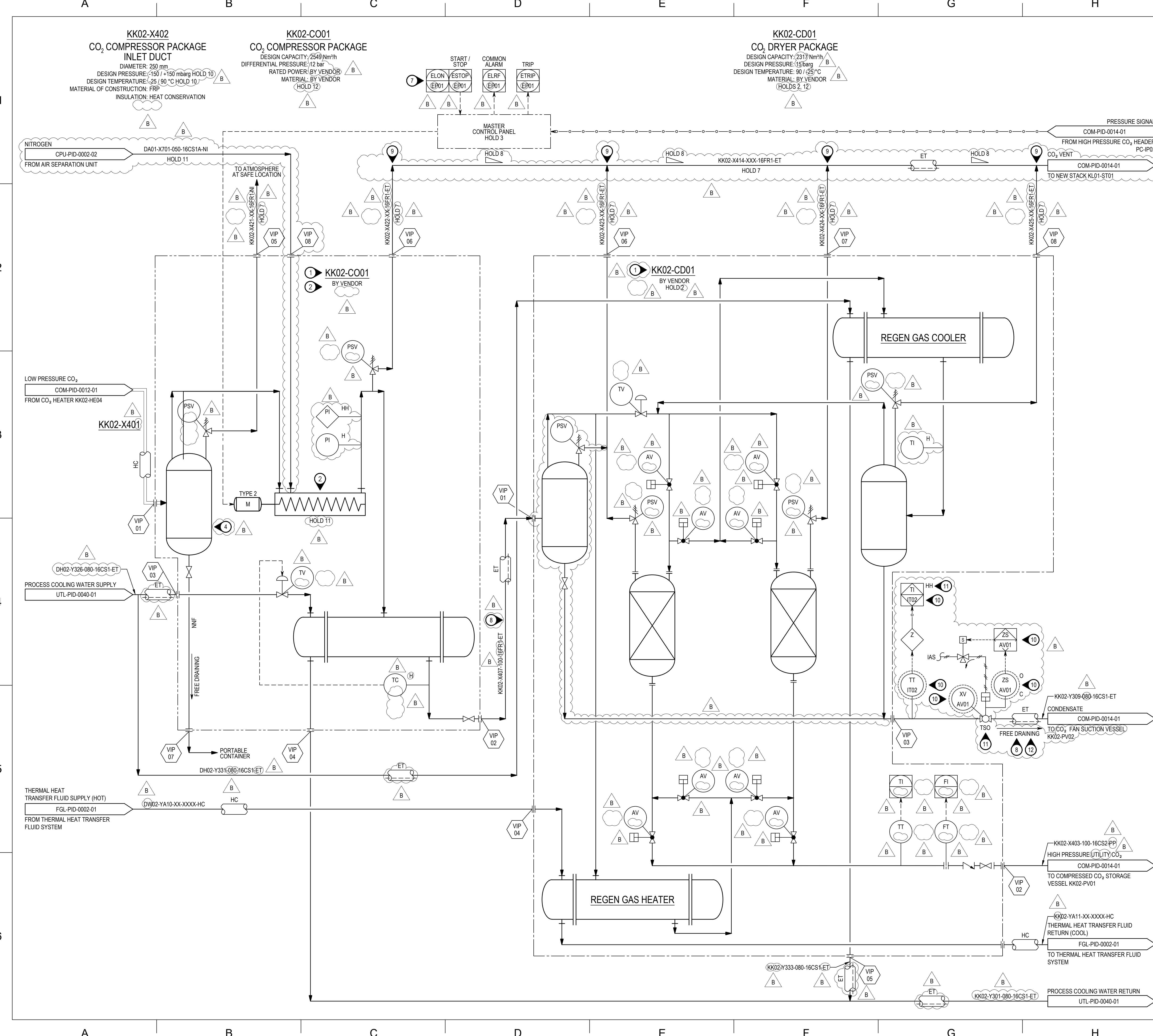
REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT
B	12MAR2025	IFH	ISSUED FOR HAZOP	AD	HCH	HCH/RS	KM
A	19DEC2024	IFR	ISSUED FOR CLIENT REVIEW	EA	HCH	HCH/RS	

	GEZERO PROJECT	
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PIPING AND INSTRUMENTATION DIAGRAM
CO₂ NETWORK
CO₂ FAN AND CO₂ HEATER

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-COM-225-PID-001-0012-01	A1	B

FILE NAME: G3ZR-COM-225-PID-001-0012-01_B.dwg



GENERAL NOTES:

1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

1. DURING EPC INDICATIVE PACKAGE DETAILS WILL BE REPLACED BY VENDOR P&IDS.

2. CO₂ COMPRESSOR PACKAGE IS EXPECTED TO INCLUDE SUCTION DRUM, 1 x 100% 2-STAGE SCREW-TYPE COMPRESSOR (HOLD 9) INTERCOOLER, 2ND STAGE SUCTION DRUM, AND AFTERCOOLER (HOLD 10).

3. DELETED.

4. VESSEL IS EXPECTED TO BE DRY.

5. 10mm WEEP HOLE TO BE PROVIDED AT LOWEST POINT OF PSV OUTLET / VENT LINE.

6. DELETED.

7. TAGS PREFIXED WITH PANEL NUMBER.

8. TWO PHASE FLOW ANCHOR SECURELY. VERTICAL RISERS SHALL BE MINIMISED AND SHALL NOT BE POCKETED.

9. PSV TAIL PIPE TO BE TOP ENTRY INTO CO₂ VENT HEADER.

10. TAGS PREFIXED WITH KK01_CD01.

11. XV TO CLOSE ON HIGH HIGH TEMPERATURE TO PROTECT DOWNSTREAM SYSTEM WHICH IS DESIGNED FOR 90 °C.

12. TO BE FREE DRAINING FROM LEVEL CONTROL VALVE IN PACKAGE (I.E. ENTIRE PACKAGE DOES NOT NEED TO BE ELEVATED).

HOLDS:

1. DUCT DESIGN CONDITIONS TO BE CONFIRMED BY THIRD PARTY (GEA).

2. TYPE OF CO₂ DRYER AND REGENERATION METHOD TO BE CONFIRMED BY VENDOR.

3. CONTROL AND INSTRUMENTATION INTERFACE WITH PACKAGE VENDOR TO BE CONFIRMED.

4. DELETED.

5. DELETED.

6. DELETED.

7. RELIEF VALVE TAIL PIPES AND CO₂ VENT HEADER LINE SIZES.

8. REQUIREMENT FOR LIQUID HANDLING FACILITY AND SLOPES IN THE CO₂ VENTS PIPING / DUCTING.

9. COMPRESSOR TYPE. IF SCREW TYPE COMPRESSOR IS SELECTED, THERE MAY BE A REQUIREMENT FOR WATER INJECTION TO REDUCE THE INTERNAL TEMPERATURE.

10. CONTENTS OF VENDOR PACKAGE.

11. COMPRESSOR BUFFER FLUID.

12. EQUIPMENT TRIM MATERIAL SPECIFICATION BY VENDOR.

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B	12MAR2025	IFH	ISSUED FOR HAZOP	AD	HCH	HCH/RS	KM
A	19DEC2024	IFR	ISSUED FOR CLIENT REVIEW	EA	HCH	HCH/RS	
REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT

GEZERO PROJECT

PIPING AND INSTRUMENTATION DIAGRAM
CO₂ NETWORK
CO₂ COMPRESSOR AND CO₂ DRYER PACKAGES

SCALE
N/A

DRAWING NUMBER
G3ZR-COM-225-PID-001-0013-01

SIZE
A1

REVISION
B

FILE NAME: G3ZR-COM-225-PID-001-0013-01_B.dwg

- GENERAL NOTES:
1.

FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS
G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.
- NOTES:
1.

ANALYSER TO BE LOCATED CLOSE TO CO₂ DRYER PACKAGE (KK02-CD01) DISCHARGE. HIGH PRESSURE DRY CO₂ IS DEW POINTED TO - 40 °C @ 8 BARG IN THE CO₂ DRYER PACKAGE (KK02-CD01).
2.

THE COMPRESSED CO₂ STORAGE VESSEL IS EXPECTED TO BE DRY. THEREFORE, UNDER NORMAL OPERATING CONDITIONS THERE IS NO LIQUID LEVEL AND NO FLOW OF CONDENSATE.
3.

DELETED.
4.

DELETED.
5.

TAGS PREFIXED WITH KK02_CD01.
6.

TAGS PREFIXED WITH KK02_PV01.
7.

ANALYSER REQUIRED TO ENSURE THERE IS NO LIQUID WEATHERING IN THE COMPRESSED CO₂ STORAGE VESSEL (KK02-PV01).
8.

KEY OPERATED INTERLOCK SYSTEM.
9.

PSV TAIL PIPE TO BE TOP ENTRY INTO CO₂ VENT HEADER.
10.

TWO PHASE FLOW ANCHOR SECURELY. VERTICAL RISERS SHALL BE MINIMISED AND SHALL NOT BE POCKETED.
11.

COMPRESSED CO₂ STORAGE VESSEL TO BE ELECTRICALLY TRACED UP TO HIGH LIQUID LEVEL.

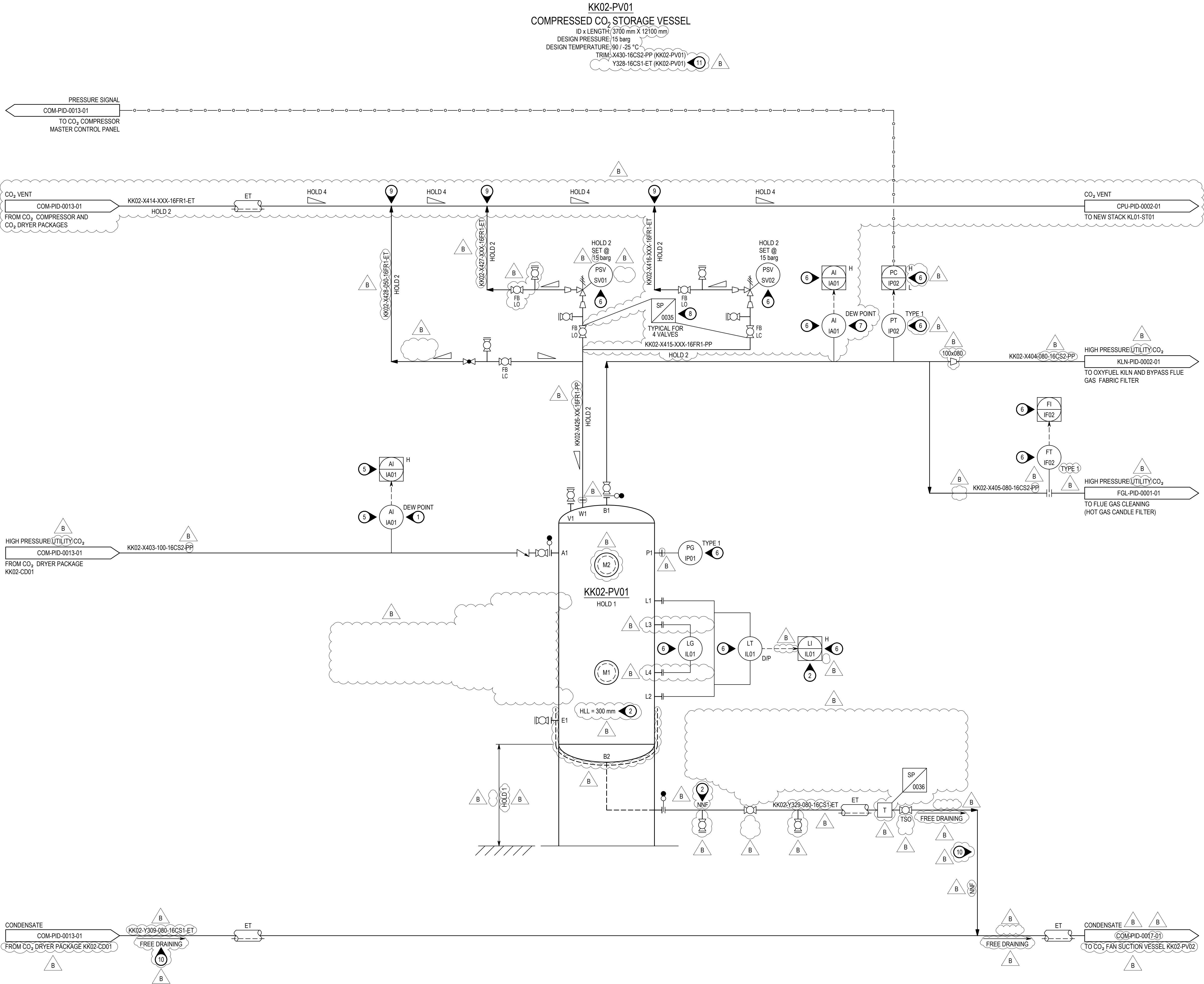
- HOLDS:
1.

VESSEL ELEVATION TO ALLOW FOR FREE DRAINING OF CONDENSATE TO CO₂ FAN SUCTION VESSEL FROM DRAIN TRAP.
2.

RELIEF VALVE SIZE, INLET PIPING, TAIL PIPES AND CO₂ VENT HEADER LINE SIZES.
3.

DELETED.
4.

REQUIREMENT FOR LIQUID HANDLING FACILITY AND SLOPES IN THE CO₂ VENTS PIPING / DUCTING.



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A	19DEC2024	IFR	ISSUED FOR CLIENT REVIEW	EA	HCH	HCH/RS	



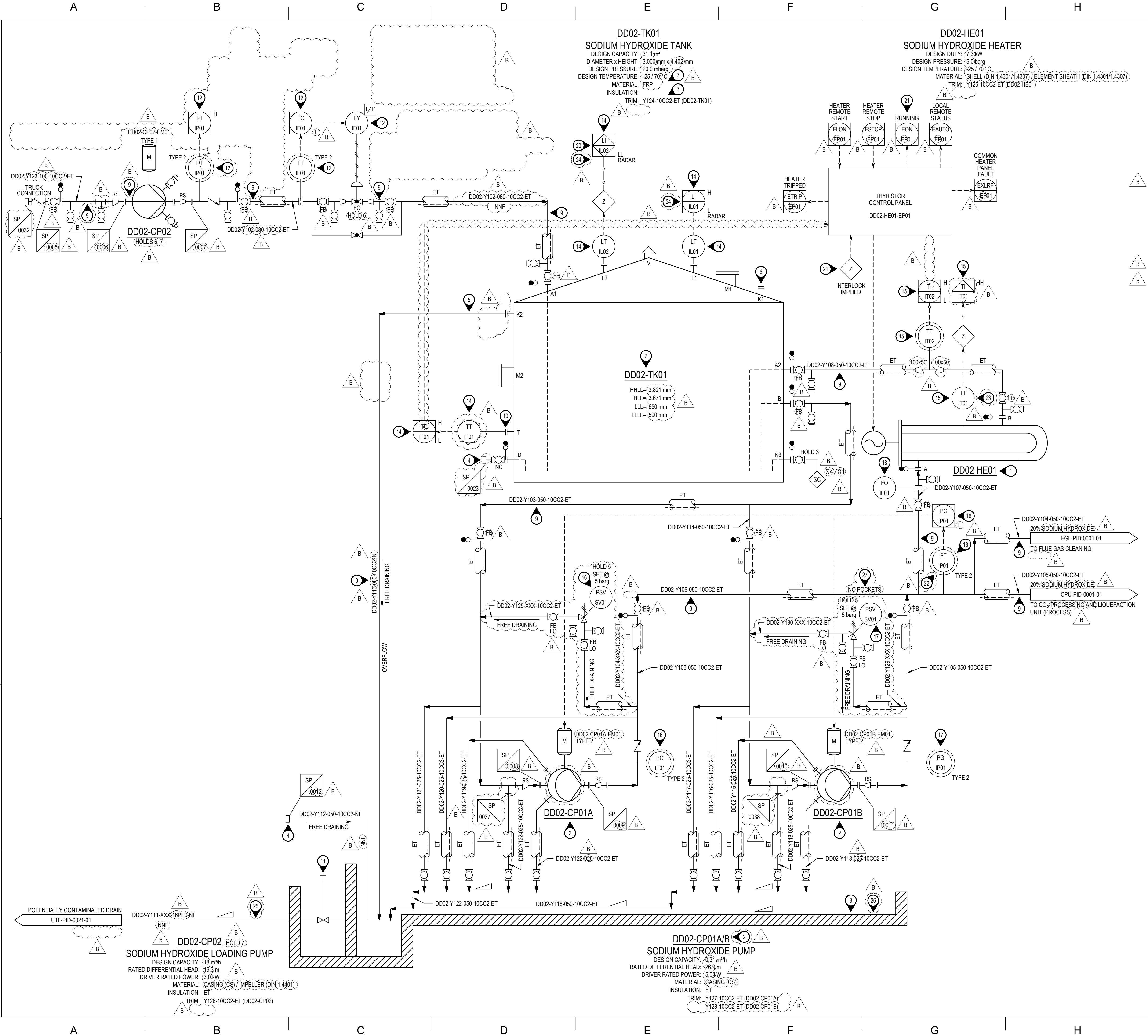
GEZERO PROJECT



PIPING AND INSTRUMENTATION DIAGRAM
CO₂ NETWORK
COMPRESSED CO₂ STORAGE VESSEL AND DISTRIBUTION

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-COM-225-PID-001-0014-01	A1	B

FILE NAME: G3ZR-COM-225-PID-001-0014-01_B.dwg



GENERAL NOTES:

- FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

- EXTERNAL HEATER TO BE INSTALLED TO MAINTAIN TEMPERATURE OF 20 WT% SODIUM HYDROXIDE IN THE SODIUM HYDROXIDE TANK AT 5 °C.
- 2 x 100% MOTOR DRIVEN DIAPHRAGM PUMP. DATA SHOWN IS FOR A SINGLE PUMP.
- CURB / BUND AREA REQUIRES CAUSTIC ALKALI RESISTANT COATING.
- VACUUM TRUCK CONNECTION.
- THE OVERFLOW / DRAIN LINE SHALL BE EXTENDED AS CLOSE AS POSSIBLE TO THE FLOOR OF THE CURB TO AVOID SPLASHING.
- GAUGING HATCH.
- SODIUM HYDROXIDE TANK TO BE A SINGLE-WALLED FRP TANK, EXTERNALLY INSULATED WITH INTERNAL LAMINATE WHICH IS CORROSIVE RESISTANT AGAINST THE STORED FLUID.
- DELETED.
- HEAT TRACE TO BE APPLIED TO MAINTAIN TEMPERATURE OF 20WT % SODIUM HYDROXIDE AT 5 °C.
- TT-IT01 TO BE LOCATED BELOW LOW LOW LIQUID LEVEL.
- VALVE HANDWHEEL TO EXTEND ABOVE GRADE.
- TAGS PREFIXED DD02-CP02.
- TAGS PREFIXED GW02-WP01.
- TAGS PREFIXED DD02-TK01.
- TAGS PREFIXED DD02-HE01.
- TAGS PREFIXED DD02-CP01A.
- TAGS PREFIXED DD02-CP01B.
- TAGS PREFIXED DD02-CP01.
- DELETED.
- LOW LOW LEVEL IN DD02-TK01 TO TRIP SODIUM HYDROXIDE PUMPS DD02-CP01A/B (AND SODIUM HYDROXIDE HEATER DD02-HE01).
- TAGS PREFIXED WITH PANEL NUMBER.
- LOW PRESSURE TO START STANDBY PUMP AND TO STOP OPERATING PUMP.
- TEMPERATURE TRANSMITTER MEASURES THE FLUID TEMPERATURE IN THE SHELL OF THE HEATER.
- LEVEL ALARMS TO BE LOCATED OUTSIDE LIQUID LEVEL THRESHOLDS FOR THE ON / OFF CONTROL.
- UNDERGROUND LINE BY CIVIL.
- CURB IS DEDICATED FOR ALL EQUIPMENT IN THE SODIUM HYDROXIDE STORAGE AND PUMPS AREA.
- LINE TO HAVE NO POCKETS TO ENABLE DRAINING TO EITHER SOURCE OR DESTINATION.

HOLDS:

- DELETED.
- DELETED.
- SAMPLING CONNECTION AND LOCATION.
- DELETED.
- RELIEF VALVE SIZE AND LINE SIZE.
- EPC CONTRACTOR TO OPTIMISE THE CONTROL SCHEME ONCE PUMP INFORMATION IS AVAILABLE (SUCH AS POTENTIAL TO REMOVE FLOW CONTROL VALVE ON 20 WT% SODIUM HYDROXIDE LOADING LINE).
- TYPE OF PUMP TO MEET SELF-PRIMING AND MINIMUM FLOW REQUIREMENT.

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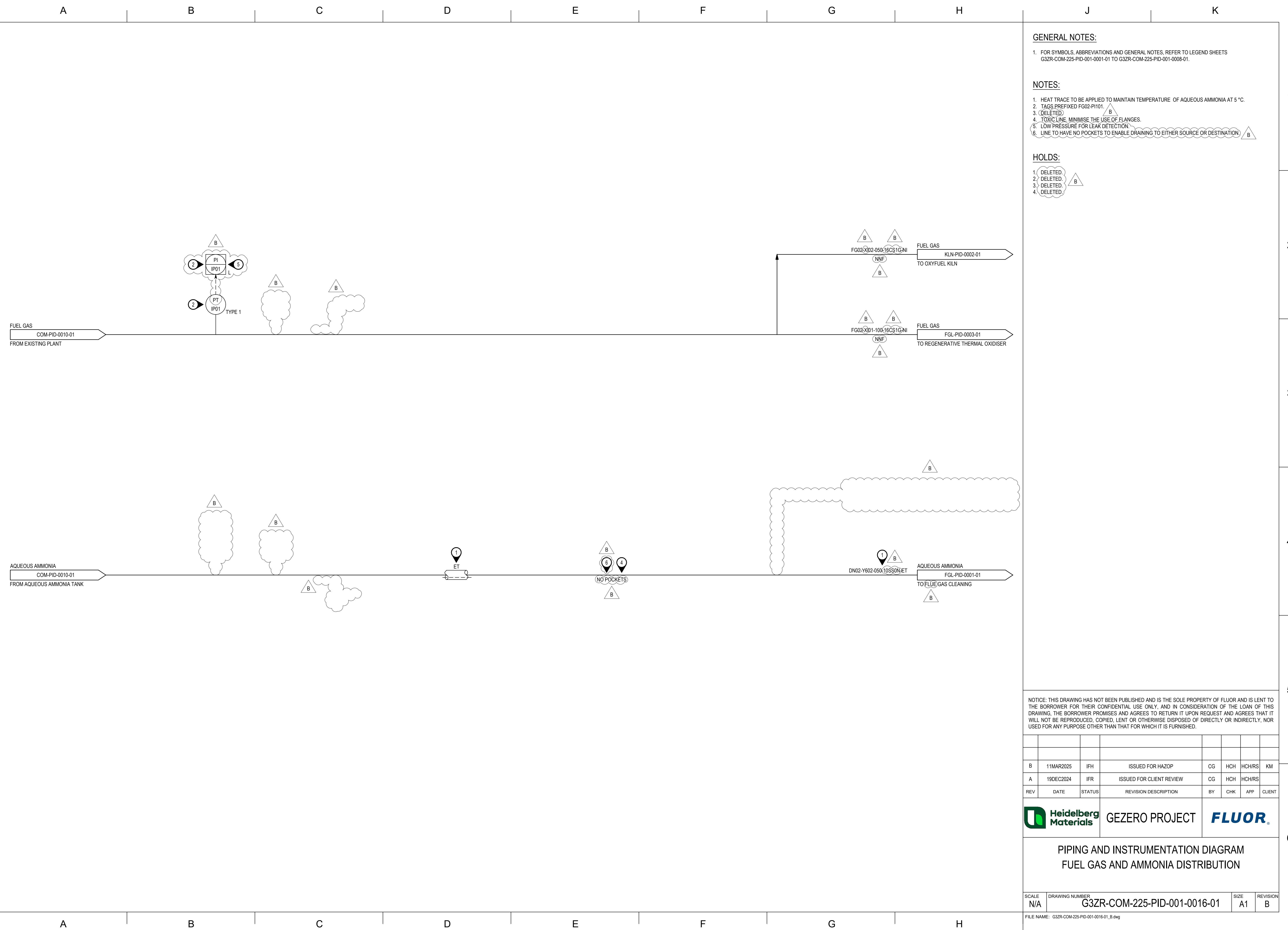
REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT
B	12MAR2025	IFH	ISSUED FOR HAZOP	CG	HCH	HCH/RS	KM
A	19DEC2024	IFR	ISSUED FOR CLIENT REVIEW	CG	HCH	HCH/RS	

Heidelberg Materials	GEZERO PROJECT	FLUOR
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PIPING AND INSTRUMENTATION DIAGRAM SODIUM HYDROXIDE STORAGE TANK AND PUMP

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-COM-225-PID-001-0015-01	A1	B

FILE NAME: G3ZR-COM-225-PID-001-0015-01_B.dwg



GENERAL NOTES

1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS
G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

1. HEAT TRACE TO BE APPLIED TO MAINTAIN TEMPERATURE OF AQUEOUS AMMONIA AT 5 °C.
2. TAGS PREFIXED FG02-P1101.
3.  (DELETED)
4. TOXIC LINE, MINIMISE THE USE OF FLANGES.
5. LOW PRESSURE FOR LEAK DETECTION.
6. LINE TO HAVE NO POCKETS TO ENABLE DRAINING TO EITHER SOURCE OR DESTINATION.

HOLDS:

1. DELETED.
2. DELETED.
3. DELETED.
4. DELETED.
-  B

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

B	11MAR2025	IFH	ISSUED FOR HAZOP	CG	HCH	HCH/RS	KM
A	19DEC2024	IFR	ISSUED FOR CLIENT REVIEW	CG	HCH	HCH/RS	
REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT



GEZERO PROJECT

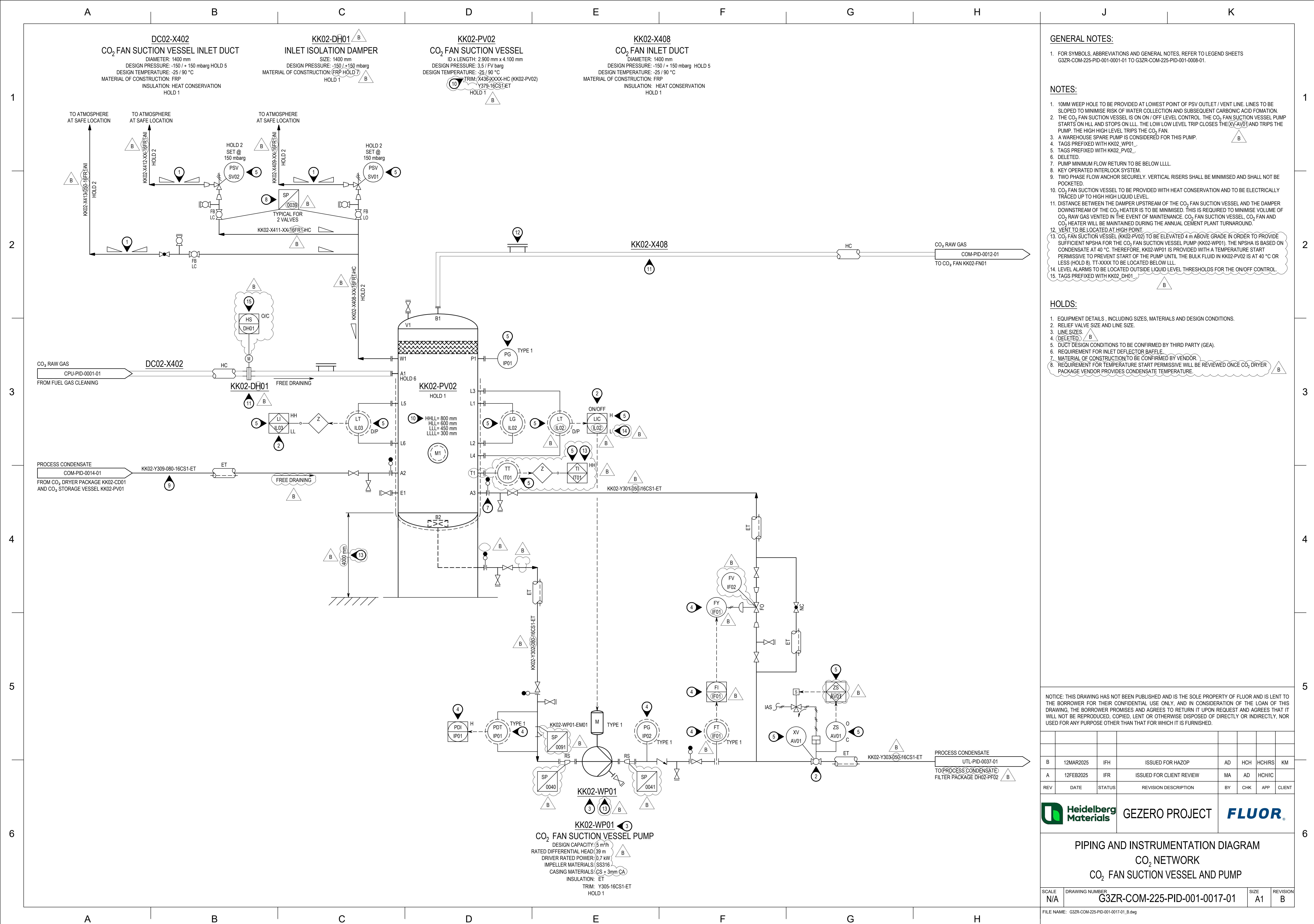


PIPING AND INSTRUMENTATION DIAGRAM

FUEL GAS AND AMMONIA DISTRIBUTION

SCALE N/A	DRAWING NUMBER G3ZR-COM-225-PID-001-0016-01	SIZE A1	REVISION B
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FILE NAME: G3ZR-COM-225-PID-001-0016-01_B.dwg



GENERAL NOTES:

1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS
G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

1. PACKAGE ISOLATION IS WITHIN THIRD PARTY (LINDE) SCOPE.
2. (HEAT TRACING TO BE APPLIED TO MAINTAIN TEMPERATURE OF 20 WT% SODIUM HYDROXIDE AT 5 °C.
3. PROCESS CONDENSATE FLASH DRUM TO BE PROVIDED IN THIRD PARTY (LINDE) SCOPE.
4. (DUCTING / PIPING TO HAVE NO POCKETS TO ENABLE DRAINING TO EITHER SOURCE OR DESTINATION.

HOLDS:

1. THIRD PARTY P&ID NUMBER(S).
2. (DELETED).
3. REQUIREMENT FOR DUCTING VERSUS PIPING.
4. (DELETED).
5. LINE AND DUCT SIZING TO BE CONFIRMED BASED ON THIRD PARTY (GEA / LINDE) PROCESS DATA.
6. DUCT DESIGN CONDITIONS TO BE CONFIRMED BASED ON THIRD PARTY (GEA / LINDE) PROCESS DATA.
7. EPC CONTRACTOR TO PROVIDE HIGH PONT VENTS AND LOW POINT DRAINS IN-LINE WITH THIRD PARTY (LINDE-CPU) DRAINING / VENTING PHILOSOPHY.
8. EPC CONTRACTOR TO CONVERT PIPING MATERIAL CLASSES PROVIDED BY THIRD PARTIES TO A CONSISTENT PROJECT PIPING CLASS INDEX AND RESOLVE ANY DISCREPANCIES.
9. MATERIAL OF CONSTRUCTION.

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY. AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT
B	12MAR2025	IFH	ISSUED FOR HAZOP	AD	HCH	HCH/RS	KM
A	19DEC2024	IFR	ISSUED FOR CLIENT REVIEW	AG	HCH	HCH/RS	



GEZERO PROJECT



PIPING AND INSTRUMENTATION DIAGRAM
CO₂ PROCESSING AND LIQUIFACTION UNIT (PROCESS)

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-CPU-225-PID-001-0001-01	A1	B

FILE NAME: G3ZR-CPU-225-PID-001-0001-01_B.dwg

GENERAL NOTES:

1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS
G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

1. PACKAGE ISOLATION BY THIRD PARTY (FLUE GAS CLEANING).
2. ALL THERMAL FLUID LINES AND EQUIPMENT TO BE FREE DRAINING TO COLLECTION VESSELS FROM HIGH POINT (HOLD 5).

HOLDS:

1. THIRD PARTY P&ID NUMBER(S).
2. (DELETED).
3. THERMAL HEAT TRANSFER FLUID PIPING SPECIFICATION AND SIZES BY THIRD PARTY (FLUE GAS CLEANING).
4. CONFIGURATION OF THERMAL FLUID SYSTEM.
5. LOCATION OF THERMAL FLUID DRAIN VESSELS AND THEREFORE PIPING HIGH POINT LOCATIONS.
6. EPC CONTRACTOR TO CONVERT PIPING MATERIAL CLASSES PROVIDED BY THIRD PARTIES TO A CONSISTENT PROJECT PIPING CLASS INDEX AND RESOLVE ANY DISCREPANCIES.

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

B	11MAR2025	IFH	ISSUED FOR HAZOP	PB	HCH	HCH/RS	KM
A	19DEC2024	IFR	ISSUED FOR CLIENT REVIEW	PB	HCH	HCH/RS	
REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT



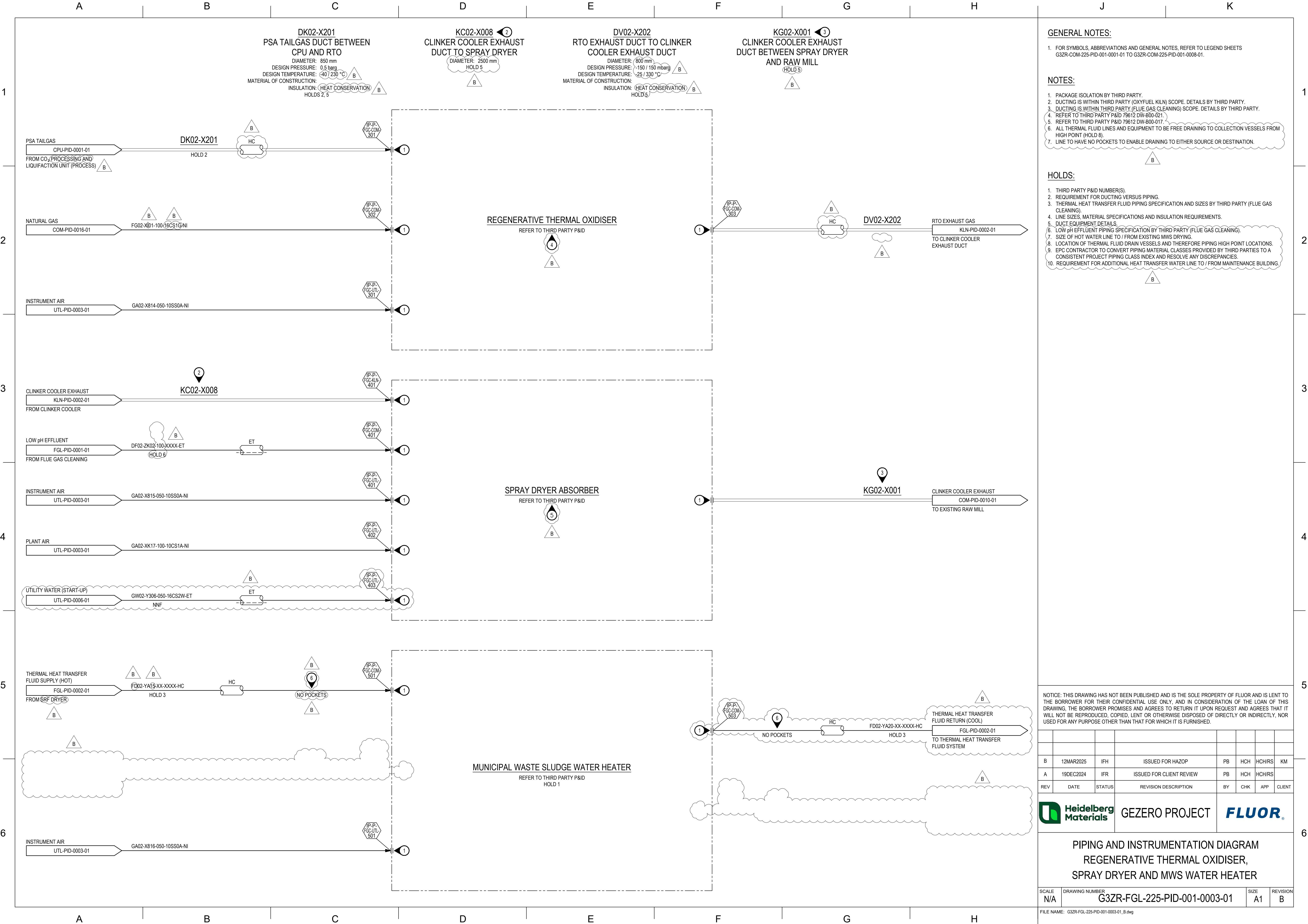
GEZERO PROJECT

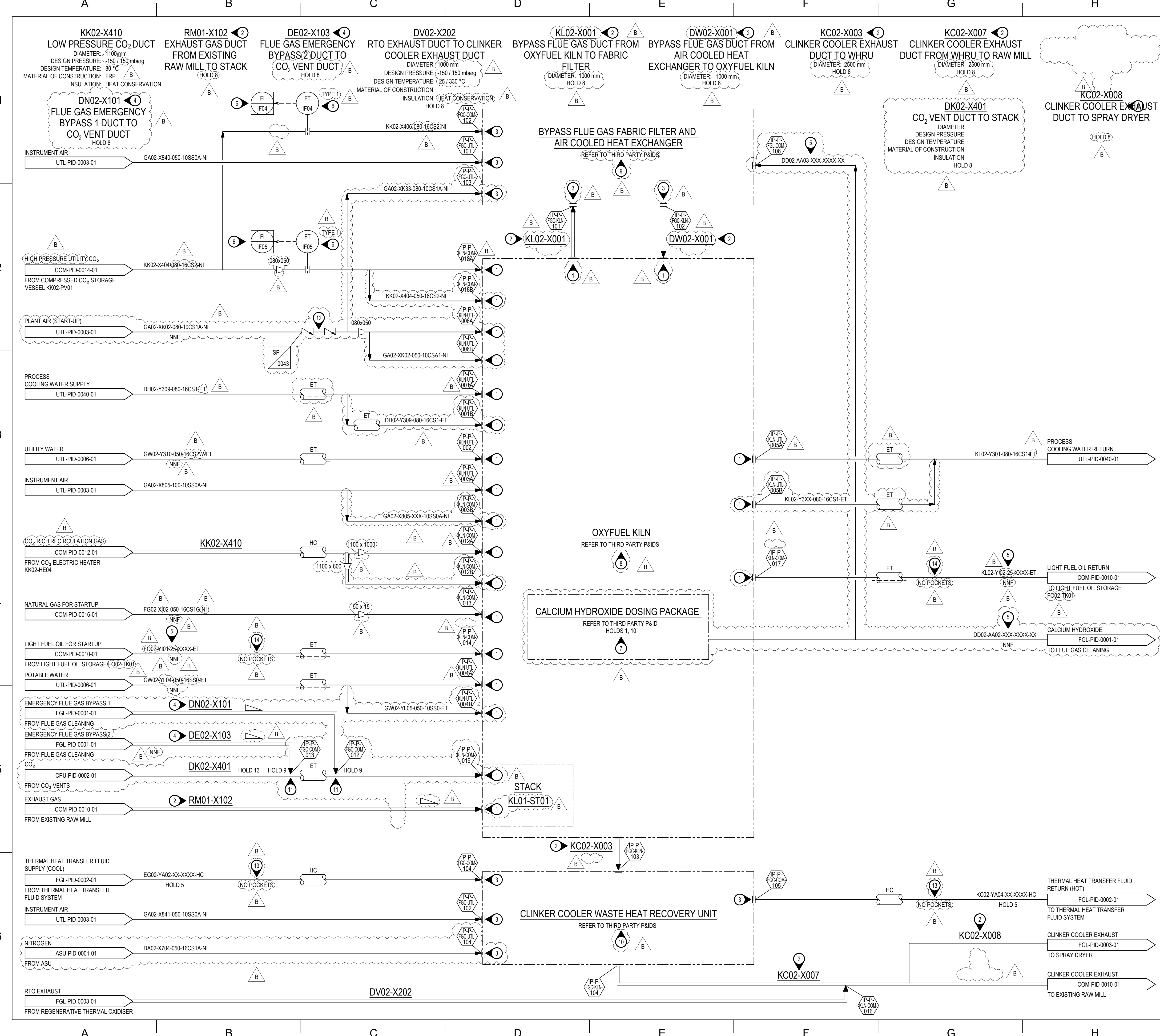


PIPING AND INSTRUMENTATION DIAGRAM
THERMAL HEAT TRANSFER FLUID SYSTEM

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-FGL-225-PID-001-0002-01	A1	B

FILE NAME: G3ZR-FGL-225-PID-001-0002-01_B.dwg





- GENERAL NOTES:**
- FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.
- NOTES:**
- PACKAGE ISOLATION IS WITHIN THIRD PARTY (OXYFUEL KILN) SCOPE.
 - DUCTING IS WITHIN THIRD PARTY (OXYFUEL KILN) SCOPE. DETAILS BY THIRD PARTY.
 - PACKAGE ISOLATIONS 1 WITHIN THIRD PARTY (FLUE GAS CLEANING) SCOPE.
 - DUCTING IS WITHIN THIRD PARTY (FLUE GAS CLEANING) SCOPE. DETAILS BY THIRD PARTY.
 - PIPING IS WITHIN HM SCOPE.
 - TAGS PREFIXED WITH KK02-PV01.
 - CALCIUM HYDROXIDE DOSING PACKAGE TO BE PROVIDED WITH INSTRUMENT AIR FROM WITHIN OXYFUEL KILN BATTERY LIMIT (HOLD 10).
 - REFER TO THIRD PARTY (OXYFUEL KILN) P&IDS A-05-0005 SHEET 1 TO 31.
 - REFER TO THIRD PARTY (FLUE GAS CLEANING) P&IDS 79612-DW-B00-019, -020.
 - REFER TO THIRD PARTY (FLUE GAS CLEANING) P&ID 79612-DW-B00-018.
 - TOP ENTRY INTO DUCT.
 - DISSIMILAR CHECK VALVE TYPES.
 - ALL THERMAL FLUID LINES AND EQUIPMENT TO BE FREE DRAINING TO COLLECTION VESSELS FROM HIGH POINT (HOLD 11).
 - LINE TO HAVE NO POCKETS TO ENABLE DRAINING TO EITHER SOURCE OR DESTINATION.

- HOLDS:**
- THIRD PARTY P&ID NUMBER(S).
 - (DELETED)
 - (DELETED)
 - (DELETED)
 - THERMAL HEAT TRANSFER FLUID PIPING SPECIFICATION AND SIZES BY THIRD PARTY (FLUE GAS CLEANING).
 - LINE SIZES, MATERIAL SPECIFICATIONS AND INSULATION REQUIREMENTS.
 - (DELETED)
 - DUCT EQUIPMENT DETAILS.
 - FLUE GAS CLEANING BYPASS DUCTS AND CO2 VENT DUCT CONFIGURATION.
 - LOCATION OF HYDRATED LIME PACKAGE.
 - LOCATION OF THERMAL FLUID DRAIN VESSELS AND THEREFORE PIPING HIGH POINT LOCATIONS.
 - EPC CONTRACTOR TO CONVERT PIPING MATERIAL CLASSES PROVIDED BY THIRD PARTIES TO A CONSISTENT PROJECT PIPING CLASS INDEX AND RESOLVE ANY DISCREPANCIES.
 - REQUIREMENT FOR LIQUID HANDLING FACILITY AND SLOPES IN THE CO2 VENTS PIPING / DUCTING

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

B	12MAR2025	IFH	ISSUED FOR HAZOP	PB	HCH	HCH/RS	KM
A	19DEC2024	IFR	ISSUED FOR CLIENT REVIEW	PB	HCH	HCH/RS	
REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT



GEZERO PROJECT



PIPING AND INSTRUMENTATION DIAGRAM
OXYFUEL KILN INTERFACES 2

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-KLN-225-PID-001-0002-01	A1	B

FILE NAME: G3ZR-KLN-225-PID-001-0002-01_B.dwg

GENERAL NOTES:

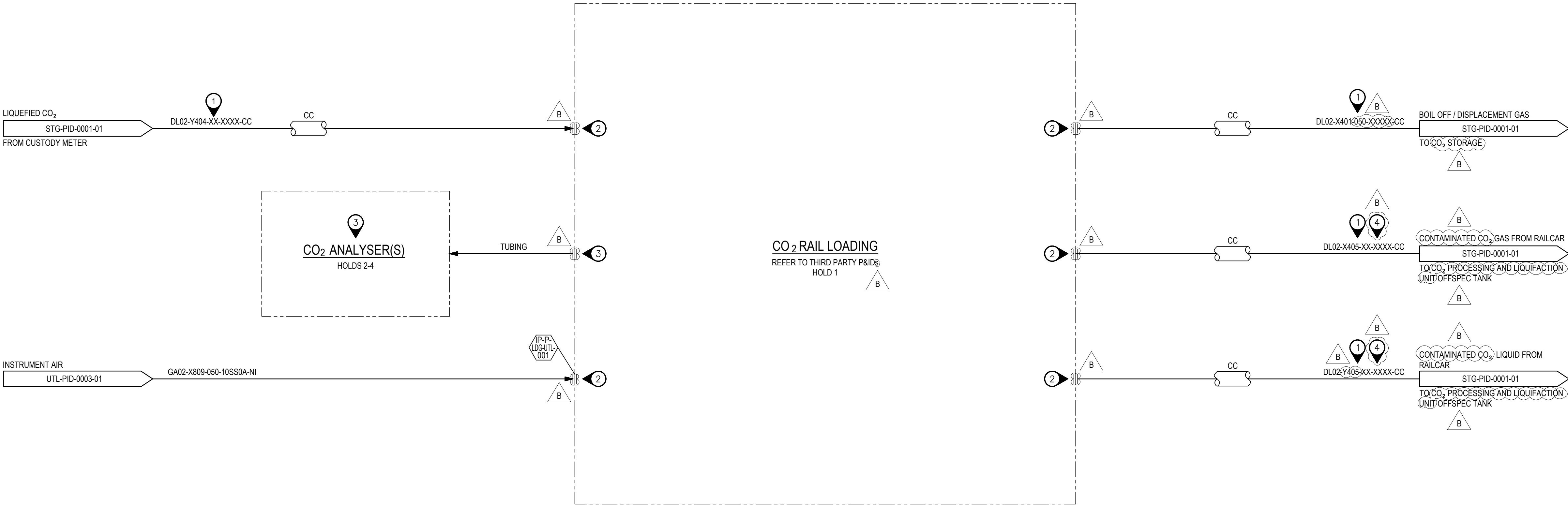
1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS
G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

1. LINE IS WITHIN THIRD PARTY (RAIL LOADING) SCOPE.
2. PACKAGE ISOLATION IS WITHIN THIRD PARTY (RAIL LOADING) SCOPE.
3. LOADING ARMS WILL BE PROVIDED WITH SAMPLING POINTS (OWNED AND SPECIFIED BY LINDE) TO CONFIRM CO₂ QUALITY. RAIL LOADING AREA WILL ALSO INCLUDE ANALYSERS (OWNED AND SPECIFIED BY LINDE).
4. INCOMING CONTAMINATED CO₂ FROM THE RAIL CAR WILL BE MEASURED IN RAIL LOADING WEIGH STATION

HOLDS:

1. THIRD PARTY P&ID NUMBER(S).
2. NUMBER OF ANALYSERS TO BE CONFIRMED BY LINDE.
3. RETURN LOCATION OF SAMPLES FROM ANALYSERS TO BE CONFIRMED (TO MINIMISE FUGITIVE EMISSIONS).
4. DELETED.
5. THIRD PARTY LINE SIZES.
6. EPC CONTRACTOR TO CONVERT PIPING MATERIAL CLASSES PROVIDED BY THIRD PARTIES TO A CONSISTENT PROJECT PIPING CLASS INDEX AND RESOLVE ANY DISCREPANCIES.



NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT
B	12MAR2025	IFH	ISSUED FOR HAZOP	AD	HCH	HCH/RS	KM
A	18DEC2024	IFR	ISSUED FOR CLIENT REVIEW	AG	HCH	HCH/RS	



GEZERO PROJECT



PIPING AND INSTRUMENTATION DIAGRAM
CO₂ RAIL LOADING

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-LDG-225-PID-001-0001-01	A1	B

FILE NAME: G3ZR-LDG-225-PID-001-0001-01_B.dwg

GENERAL NOTES:

1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS
G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

1. LINES AND TRUCK UNLOADING PACKAGE ARE WITHIN THIRD PARTY (LINDE) SCOPE.
2. PACKAGE ISOLATION IS WITHIN THIRD PARTY (LINDE) SCOPE.
3. LINE IS WITHIN THIRD PARTY (RAIL LOADING) SCOPE.
4. CO₂ STORAGE INCLUDES EVAPORATOR FOR SUBQUALITY LIQUID CO₂.
5. CO₂ QUALIFICATION AND MEASUREMENTS FOR THIRD PARTY CERTIFICATION ONLY. CO₂ PRODUCT SPECIFICATION WILL BE CHECKED WITHIN THE CO₂ PROCESSING, LIQUIFACTION AND STORAGE AREAS (AS REQUIRED).
6. INCOMING CONTAMINATED CO₂ FROM THE RAIL CAR WILL BE MEASURED IN THE RAIL LOADING WEIGH STATION.

HOLDS:

1. THIRD PARTY P&ID NUMBER(S).
2. TRUCK UNLOADING REQUIREMENT AND DETAILS TO BE CONFIRMED.
3. THIRD PARTY LINE SIZES.
4. PROCESS FOR ACCOUNTING THE QUANTITY OF (CONTAMINATED) CO₂ BOIL OFF / DISPLACEMENT GAS FROM RAIL LOADING, AS THIS WILL REQUIRE PROCESSING WITHIN THE CO₂ PROCESSING, LIQUIFACTION AND STORAGE AREAS.
5. EPC CONTRACTOR TO PROVIDE HIGH POINT VENTS AND LOW POINT DRAINS IN-LINE WITH THIRD PARTY (LINDE-CPU) DRAWING / VENTING PHILOSOPHY.
6. EPC CONTRACTOR TO CONVERT PIPING MATERIAL CLASSES PROVIDED BY THIRD PARTIES TO A CONSISTENT PROJECT PIPING CLASS INDEX AND RESOLVE ANY DISCREPANCIES.

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT
B	12MAR2025	IFH	ISSUED FOR HAZOP	AD	HCH	HCH/RS	KM
A	19DEC2024	IFR	ISSUED FOR CLIENT REVIEW	AG	HCH	HCH/RS	



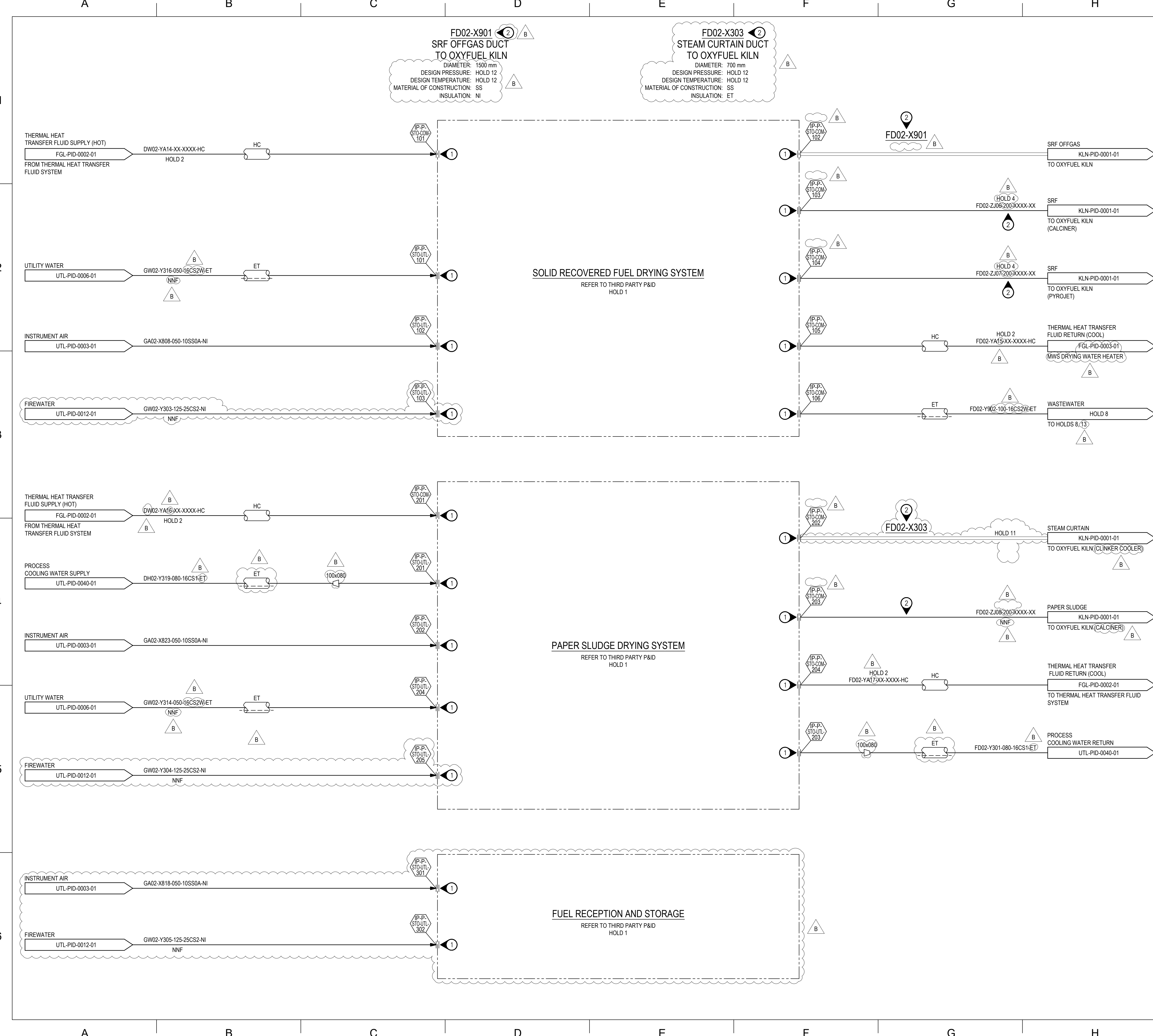
GEZERO PROJECT



PIPING AND INSTRUMENTATION DIAGRAM
CO₂ STORAGE INTERFACES

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-STG-225-PID-001-0001-01	A1	B

FILE NAME: G3ZR-STG-225-PID-001-0001-01_B.dwg



GENERAL NOTES:

- FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

- PACKAGE ISOLATION BY THIRD PARTY.
- DUCT / TRANSFER LINE IS WITHIN THIRD PARTY (ALTERNATIVE FUELS DRYING SCOPE).

HOLDS:

- THIRD PARTY P&ID NUMBER(S).
- THERMAL HEAT TRANSFER FLUID PIPING SPECIFICATION AND SIZES BY THIRD PARTY (FLUE GAS CLEANING).
- DELETED
- PIPING MATERIAL SPECIFICATION AND INSULATION BY THIRD PARTY (ALTERNATIVE FUELS DRYING).
- DELETED
- DELETED
- PIPING MATERIAL SPECIFICATIONS AND INSULATION REQUIREMENTS.
- ROUTING OF WASTEWATER.
- EPC CONTRACTOR TO CONVERT PIPING MATERIAL CLASSES PROVIDED BY THIRD PARTIES TO A CONSISTENT PROJECT PIPING CLASS INDEX AND RESOLVE ANY DISCREPANCIES.
- REQUIREMENT FOR TEMPORARY STEAM GENERATOR.
- REQUIREMENT FOR STEAM TRAPS DEPENDENT ON THE LENGTH OF LINE AND QUALITY OF STEAM.
- DUCT MECHANICAL DESIGN CONDITIONS.
- COMBINATION OF WASTEWATER FROM PAPER SLUDGE DRYING SYSTEM AND SOLID RECOVERED FUEL DRYING SYSTEM BY THIRD PARTY T BE CONFIRMED DURING EPC.

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

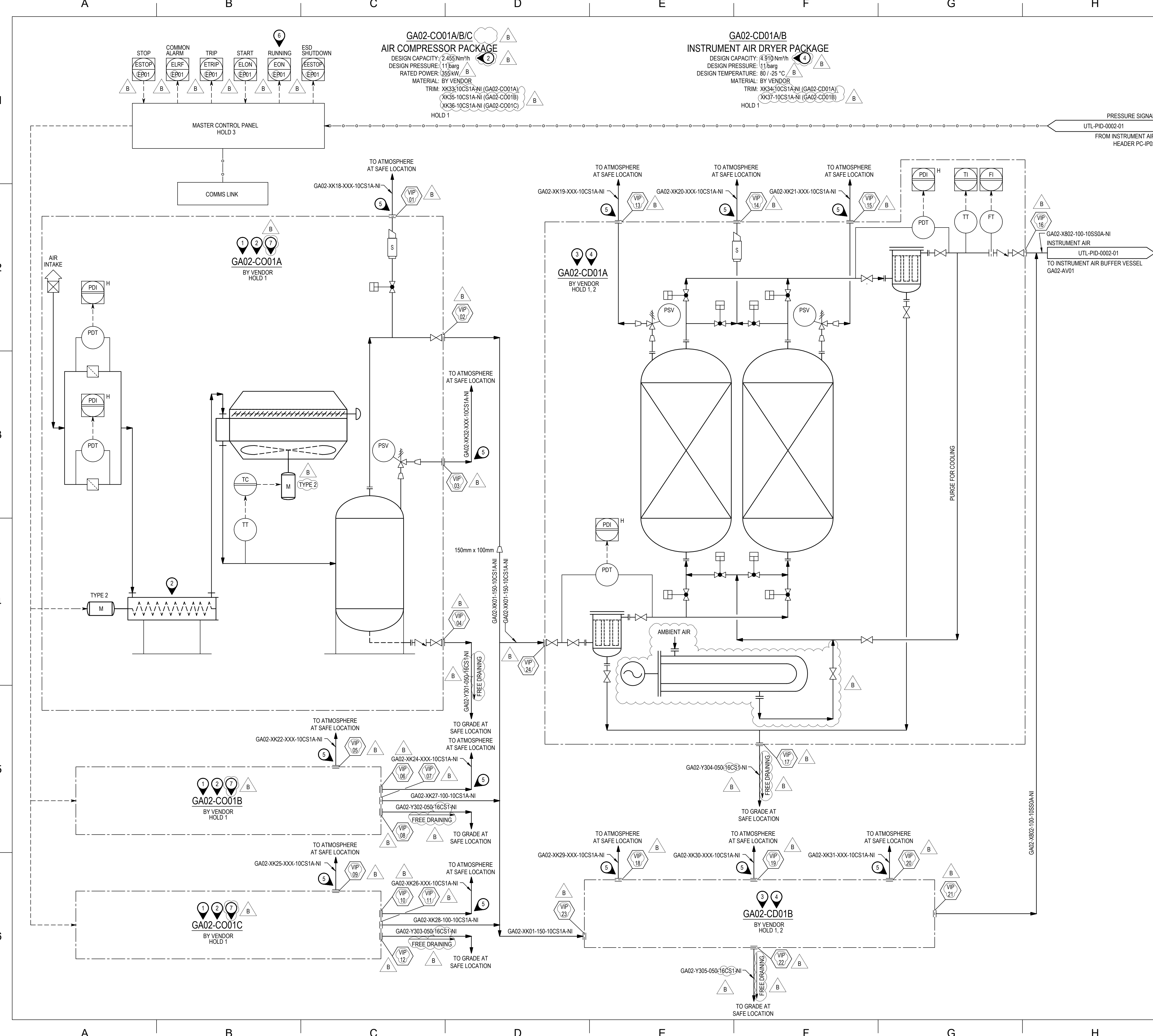
B	11MAR2025	IFH	ISSUED FOR HAZOP	PB	HCH	HCH/RS	KM
A	19DEC2024	IFR	ISSUED FOR CLIENT REVIEW	PB	HCH	HCH/RS	
REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT

GEZERO PROJECT

PIPING AND INSTRUMENTATION DIAGRAM
ALTERNATIVE FUEL DRYING SYSTEM INTERFACES

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-STO-225-PID-001-0001-01	A1	B

FILE NAME: G3ZR-STO-225-PID-001-0001-01_B.dwg



GENERAL NOTES:

1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

1. AIR COMPRESSOR IS VENDOR SUPPLY. THE EQUIPMENT INSIDE THE PACKAGE IS INDICATIVE ONLY. (DURING EPC INDICATIVE PACKAGE DETAILS WILL BE REPLACED BY VENDOR P&IDS.)
2. 3 x 50% SCREW TYPE COMPRESSORS. ALL COMPRESSOR PACKAGES ARE OF IDENTICAL DESIGN. (DATA SHOWN IS FOR ONE COMPRESSOR PACKAGE.)
3. INSTRUMENT AIR DRYER PACKAGE IS VENDOR SUPPLY. THE EQUIPMENT SHOWN INSIDE THE PACKAGE IS INDICATIVE ONLY. DURING EPC INDICATIVE PACKAGE DETAILS WILL BE REPLACED BY VENDOR P&IDS.)
4. 2 x 100% AIR DRYER TRAINS (DRYING / REGENERATION). ALL DRYER PACKAGES ARE OF IDENTICAL DESIGN. (DATA SHOWN IS FOR ONE AIR DRYER PACKAGE.)
5. 10MM WEEP HOLE TO BE PROVIDED AT LOWEST POINT OF PSV OUTLET / VENT LINE.
6. TAGS PREFIXED WITH PANEL NUMBER, PANEL NUMBER/S TO BE CONFIRMED.
7. STANDBY AIR COMPRESSOR TO BE MAINTAINED IN HOT STANDBY SO THAT IT CAN START-UP IN LESS THAN 5 MINUTES.

HOLDS:

1. EQUIPMENT DETAILS, INCLUDING SIZES AND MATERIALS.
2. (DELETED)
3. CONTROL AND INSTRUMENTATION INTERFACE WITH PACKAGE VENDOR, TO BE CONFIRMED.
4. LINE SIZE.

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

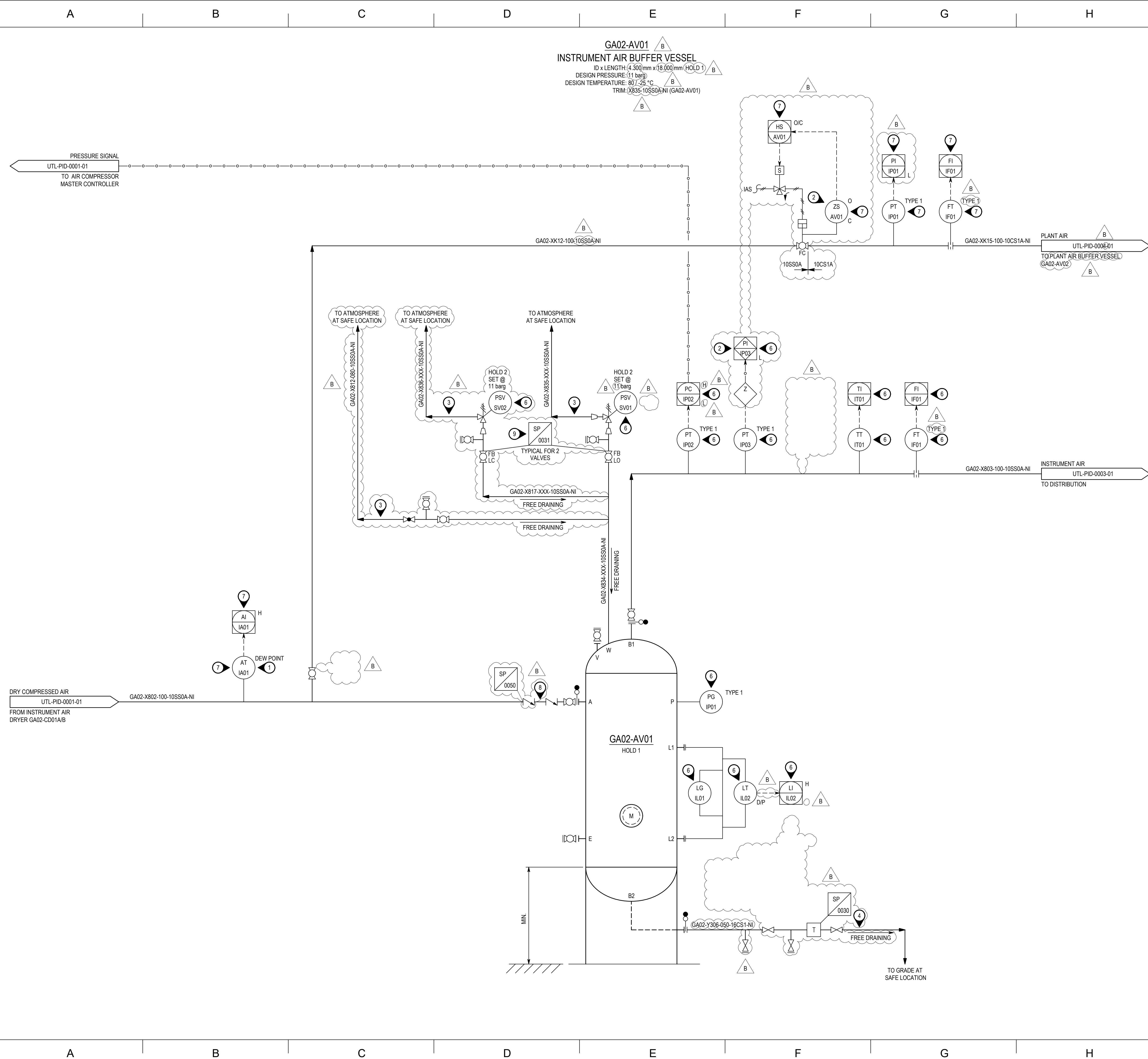
REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT
B	11MAR2025	IFH	ISSUED FOR HAZOP	CG	HCH	HCH/RS	KM
A	19DEC2024	IFR	ISSUED FOR CLIENT REVIEW	CG	HCH	HCH/RS	

Heidelberg Materials	GEZERO PROJECT	FLUOR
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PIPING AND INSTRUMENTATION DIAGRAM
AIR COMPRESSOR AND DRYER PACKAGE

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-UTL-225-PID-001-0001-01	A1	B

FILE NAME: G3ZR-UTL-225-PID-001-0001-01_B.dwg



GENERAL NOTES:

1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

1. INSTRUMENT AIR DEW POINT IS -40°C @ 8 BARG.
2. GA02-AV01-P03-10-LOW PRESSURE WILL CLOSE THE ON-OFF VALVE TO PRIORITISE INSTRUMENT AIR CONSUMERS OVER PLANT AIR IF THE INSTRUMENT HEADER PRESSURE FALLS TOO LOW.
3. 10MM WELD HOLE TO BE PROVIDED AT LOWEST POINT OF PSV OUTLET / VENT LINE.
4. THE INSTRUMENT AIR BUFFER VESSEL IS EXPECTED TO BE DRY. THEREFORE, UNDER NORMAL OPERATING CONDITIONS THERE IS NO LIQUID LEVEL AND NO FLOW OF CONDENSATE. IF THERE WILL BE LIQUID CARRY-OVER ACCUMULATED INSIDE THE VESSEL, IT WILL BE DRAINED TO GRADE THROUGH THE LIQUID TRAP.
5. ~~DELETED~~
6. TAGS PREFIXED GA02-AV01.
7. TAGS PREFIXED GA02-CD01.
8. DISSIMILAR CHECK VALVES.
9. KEY OPERATED INTERLOCK SYSTEM.

HOLDS:

1. EQUIPMENT SIZE.
2. RELIEF VALVE SIZE AND LINE SIZE.
3. LINE SIZES.

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B	11MAR2025	IFH	ISSUED FOR HAZOP		CG	HCH	HCH/RS	KM	
A	18DEC2024	IFR	ISSUED FOR CLIENT REVIEW		CG	HCH	HCH/RS		
REV	DATE	STATUS	REVISION DESCRIPTION		BY	CHK	APP	CLIENT	



GEZERO PROJECT

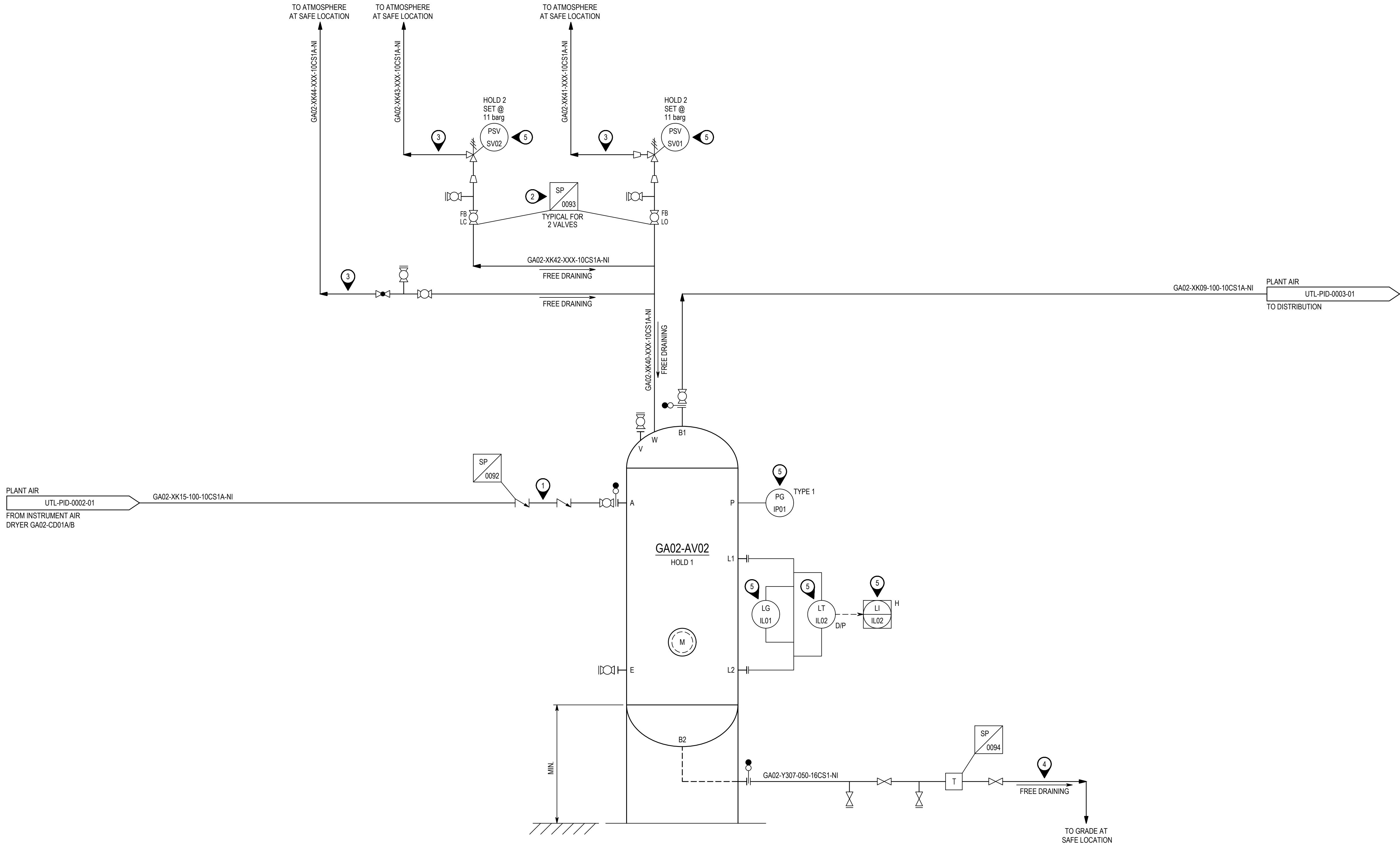


PIPING AND INSTRUMENTATION DIAGRAM INSTRUMENT AIR BUFFER VESSEL

SCALE N/A	DRAWING NUMBER G3ZR-UTL-225-PID-001-0002-01	SIZE A1	REVISION B
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FILE NAME: G3ZR-UTL-225-PID-001-0002-01_B.dwg

GA02-AV02
PLANT AIR BUFFER VESSEL
ID x LENGTH: 4.300 mm x 14.650 mm HOLD 1
DESIGN PRESSURE: 11 barg
DESIGN TEMPERATURE: 80 / -25 °C
TRIM: XK38-10CS1A-NI (GA02-AV02)



GENERAL NOTES:

- FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

- DISSIMILAR CHECK VALVES.
- KEY OPERATED INTERLOCK SYSTEM.
- 10 MM WEEP HOLE TO BE PROVIDED AT LOWEST POINT OF PSV OUTLET / VENT LINE.
- THE PLANT AIR BUFFER VESSEL IS EXPECTED TO BE DRY. THEREFORE, UNDER NORMAL OPERATING CONDITIONS THERE IS NO LIQUID LEVEL AND NO FLOW OF CONDENSATE. IF THERE WILL BE LIQUID CARRY-OVER ACCUMULATED INSIDE THE VESSEL, IT WILL BE DRAINED TO GRADE THROUGH THE LIQUID TRAP.
- TAGS PREFIXED GA02-AV02.

HOLDS:

- EQUIPMENT SIZE.
- RELIEF VALVE SIZE AND LINE SIZE.
- LINE SIZES.

THIS P&ID HAS BEEN ISSUED DIRECTLY AT IFH, AS IT IS BASED ON G3ZR-UTL-225-PID-001-0002-01 REV A.

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY. AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

A	12MAR2025	IFH	ISSUED FOR HAZOP	AG	HCH	HCH/RS	KM
REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT



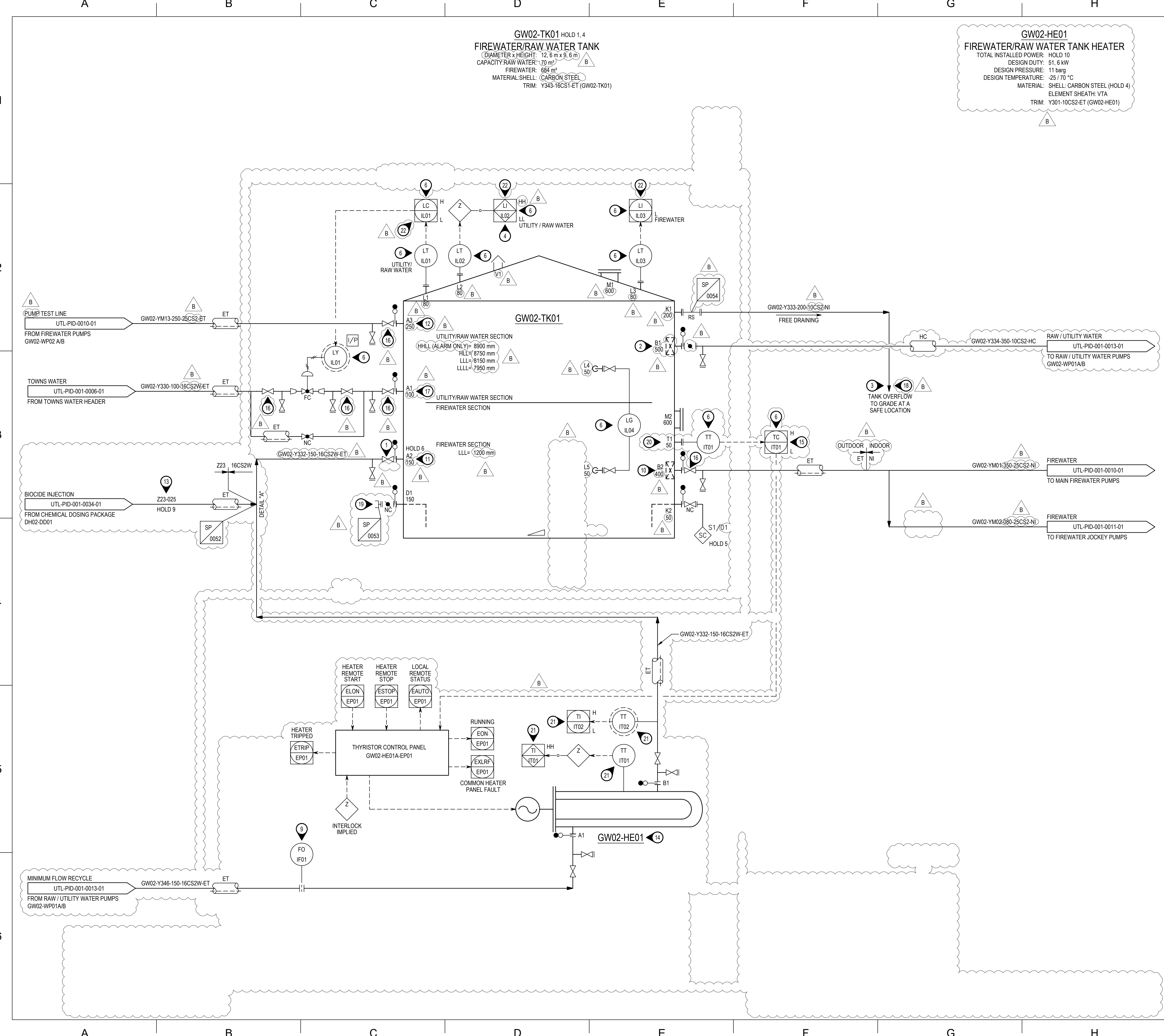
GEZERO PROJECT



PIPING AND INSTRUMENTATION DIAGRAM
PLANT AIR BUFFER VESSEL

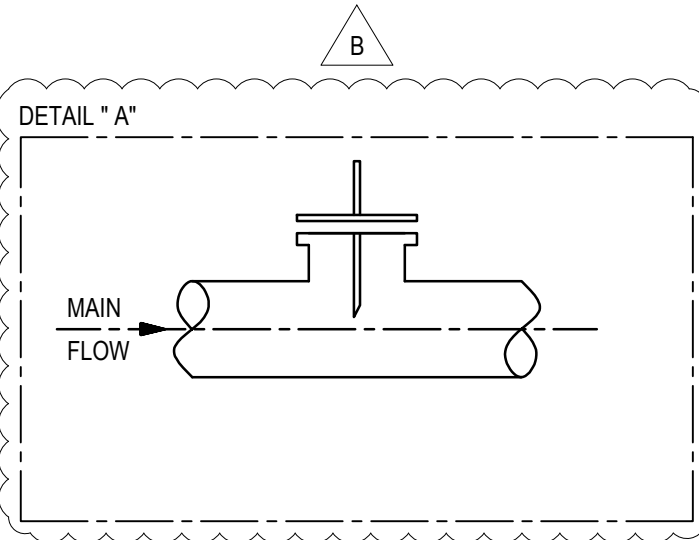
SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-UTL-225-PID-001-0004-01	A1	A

FILE NAME: G3ZR-UTL-225-PID-001-0004-01_A.dwg



- GENERAL NOTES:**
- FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.
 - PIPING DEADLEGS AND POCKETS SHALL BE ELIMINATED.
- NOTES:**
- BIOCIDE TO BE INJECTED TO PREVENT BIOLOGICAL GROWTH IN TANK (CONTINUOUS MINIMUM FLOW (PROMOTES MIXING OF BIOCIDE IN THE TANK).
 - RAW/UTILITY WATER DRAW OFF NOZZLE TO BE LOCATED ABOVE THE FIRE WATER SECTION TO ENSURE AVAILABILITY OF FIRE WATER.
 - FLAP VALVE TO BE INSTALLED AT THE BOTTOM END OF THE OVERFLOW LINE TO PREVENT ANIMALS AND INSECTS FROM ENTERING THE PIPE.
 - LOW LOW LEVEL IN RAW/UTILITY WATER SECTION TO TRIP RAW/UTILITY WATER PUMPS GW02-WP01A/B. (HIGH HIGH LEVEL TO PROVIDE INDEPENDENT ALARM ONLY).
 - DELETED.
 - TAGS PREFIXED WITH GW02-TK01.
 - DELETED.
 - DELETED.
 - TAGS PREFIXED WITH GW02-WP01.
 - VORTEX PLATE FOR PUMP SUCTION NOZZLE SHALL BE DESIGNED IN ACCORDANCE WITH NFPA 22.
 - NOZZLE TO BE LOCATED BELOW LLLL OF THE UTILITY / RAW WATER SECTION.
 - NOZZLE TO BE LOCATED ABOVE THE HIGH LIQUID LEVEL OF THE UTILITY/RAW WATER SECTION.
 - LINE TO BE RUN BY THIRD PARTY SITE CONTRACTOR.
 - EXTERNAL HEATER TO BE INSTALLED TO MAINTAIN THE TEMPERATURE OF THE WATER IN THE TANK AT OR ABOVE 5, 6 °C PER NFPA 22.
 - LOW TEMPERATURE ALARM TO BE SET AT 4, 4 °C.
 - ALL ISOLATION VALVES ON FIREWATER SERVICE LINES SHALL BE OS AND Y GATE VALVES PER NFPA 20 AND 22.
 - TANK REFILL TIME IS 12.4 HOURS FOR THE FIREWATER VOLUME BASED ON 55TPH MAXIMUM TOWNS WATER SUPPLY (HOLD 12).
 - OVERFLOW WATER TO BE CONTAINED IN SUMP TO AVOID BIOCIDE IN WATER INFILTRATING LOCAL WATER BODIES.
 - VACUUM TRUCK CONNECTION.
 - NOZZLE TO BE LOCATED BELOW LLL IN FIREWATER SECTION OF THE TANK.
 - TAGS PREFIXED WITH GW02-HE01.
 - LEVEL ALARMS TO BE LOCATED OUTSIDE LIQUID LEVEL THRESHOLDS FOR THE ON / OFF CONTROL.

- HOLDS:**
- REQUIRED FIREWATER STORAGE CAPACITY AND NUMBER OF TANKS PENDING CONFIRMATION FROM THIRD PARTY CONSULTANT.
 - DELETED.
 - DELETED.
 - EQUIPMENT DETAILS, INCLUDING SIZES, MATERIALS AND DESIGN CONDITIONS.
 - SAMPLING CONNECTION DETAILS AND LOCATION.
 - REQUIREMENT FOR SLOTTED PIPE DISTRIBUTOR TO ADEQUATELY MIX BIOCIDE INTO TANK CONTENTS.
 - DELETED.
 - DELETED.
 - MATERIAL COMPATIBILITY TO BE CONFIRMED DURING EPC.
 - TOTAL INSTALLED POWER TO BE CONFIRMED DURING EPC.
 - FIREWATER STANDARD TO BE FOLLOWED, NFPA OR V&S CEA 4001, TO BE CONFIRMED DURING EPC.
 - FIREWATER CONSULTANT TO CONFIRM ACCEPTABILITY FOR FIREWATER TANK REFILL TIME.



NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT
B	12MAR2025	IFH	ISSUED FOR HAZOP	AG	HCH	HCH/RS	KM
A	18DEC2024	IFR	ISSUED FOR CLIENT REVIEW	AG	HCH	HCH/RS	

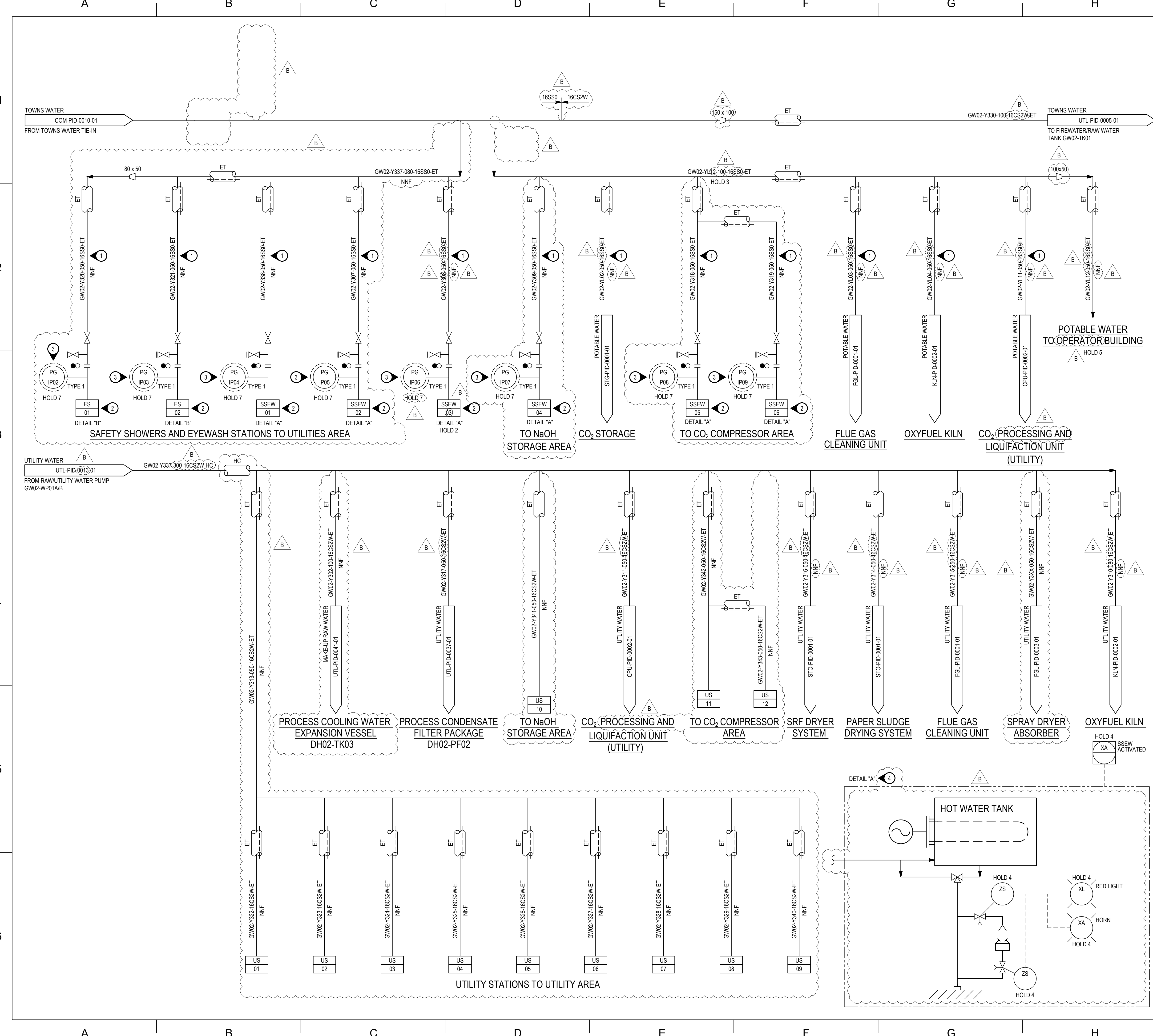
**GEZERO PROJECT**



PIPING AND INSTRUMENTATION DIAGRAM
FIREWATER/RAW WATER TANK

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-UTL-225-PID-001-0005-01	A1	B

FILE NAME: G3ZR-UTL-225-PID-001-0005-01_B.dwg



GENERAL NOTES:

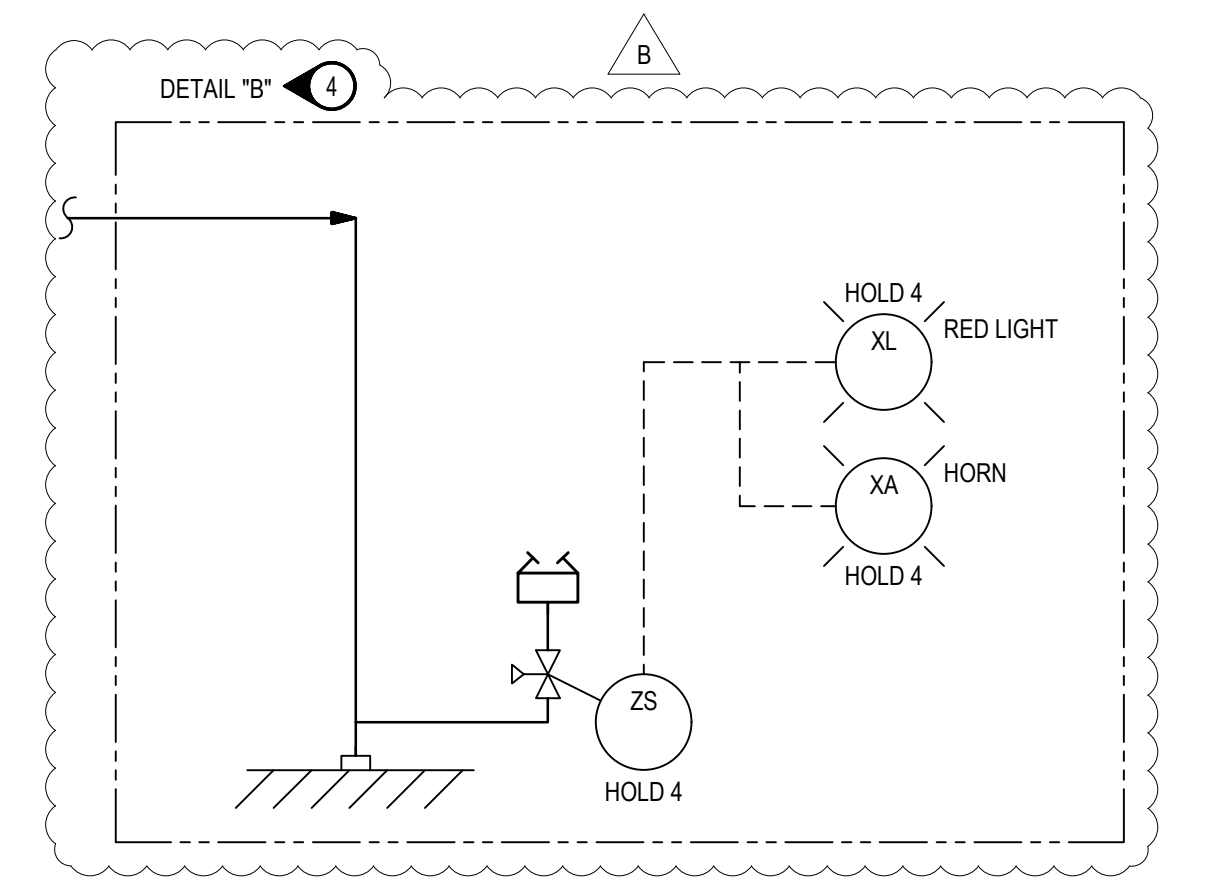
- FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

- POTABLE WATER LINES TO BE DESIGNED WITHOUT DEADLEGS.
- SAFETY SHOWER TO HEAT POTABLE WATER BETWEEN 16 - 38 °C.
- TAGS PREFIXED GW02-PID01.
- SAFETY SHOWER TO BE GRAVITY FED.

HOLDS:

- DELETED
- DELETED
- DELETED
- TAGGING OF INSTRUMENTS FOR SAFETY SHOWER / EYE WASH.
- BUILDING DRAWING NUMBER TO BE PROVIDED DURING EPC.
- DELETED
- REQUIREMENT FOR PRESSURE GAUGE TO BE CONFIRMED PENDING SAFETY SHOWER AND EYEWASH STATION PACKAGE DETAILS.



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B	12MAR2025	IFH	ISSUED FOR HAZOP	AG	HCH	HCH/RS	KM
A	19DEC2024	IFR	ISSUED FOR CLIENT REVIEW	AG	HCH	HCH/RS	

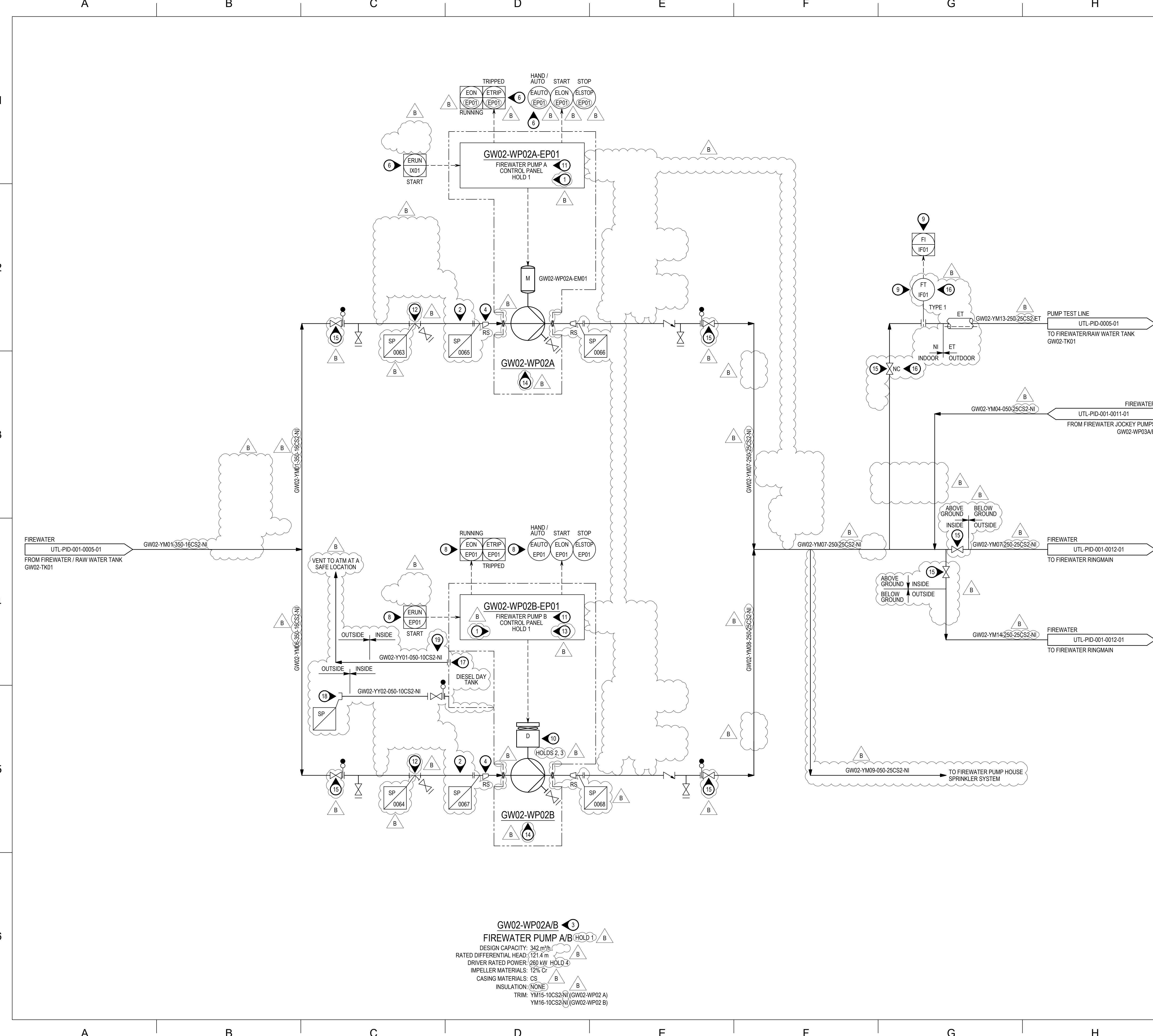
**GEZERO PROJECT**



**PIPING AND INSTRUMENTATION DIAGRAM
POTABLE AND UTILITY WATER DISTRIBUTION**

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-UTL-225-PID-001-0006-01	A1	B

FILE NAME: G3ZR-UTL-225-PID-001-0006-01_B.dwg



GENERAL NOTES:

1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

1. MAIN FIREWATER PUMPS TO START ON FAILING FIREWATER HEADER PRESSURE. PUMP START SEQUENCE TO PRIORITIZE STARTING OF ELECTRICAL DRIVEN PUMP OVER THE DIESEL DRIVEN PUMP. ONLY ONE PUMP TO OPERATE AT A TIME.

2. 10D STRAIGHT(LENGTH)PIPE FROM(PUMP)INLET PER NFPA 20.

3. MAIN FIREWATER PUMPS ARE 2X100% (ONE DUTY AND ONE STANDBY), ONE ELECTRIC DRIVER AND ONE DIESEL DRIVER. DATA SHOWN IS FOR A SINGLE PUMP.

4. ECCENTRIC REDUCER IN ACCORDANCE NFPA 20.

5. (DELETED)

6. TAGS PREFIXED GW02-WP02A.

7. (DELETED)

8. TAGS PREFIXED WITH PANEL NUMBER.

9. TAGS PREFIXED GW02-WP02.

10. DIESEL ENGINE TO INCLUDE FUEL SUPPLY TANK WITH A CAPACITY AT LEAST EQUAL TO 5.07 L PER KW. PLUS 5 PERCENT VOLUME FOR EXPANSION AND 5 PERCENT VOLUME FOR SUMP.

11. CONTROL PANELS TO BE PROVIDED BY PUMP VENDOR.

12. STRAINER FOR PUMP START-UP ONLY.

13. PUMP OVER-SPEED TRIP INTERNALLY WITHIN CONTROL PANEL.

14. PRESSURE SENSING LINE AND PRESSURE DETECTION ON SUCTION AND DISCHARGE OF THE PUMP TO BE INCLUDED AS PART OF VENDOR SCOPE OF SUPPLY.

15. ALL ISOLATION VALVES ON FIREWATER SERVICE LINES SHALL BE OS AND Y GATE VALVES PER NFPA 20 AND 22.

16. ISOLATION VALVE AND FLOW TRANSMITTER TO BE LOCALLY ACCESSIBLE AND VISIBLE FOR FIREWATER PUMP TESTING.

17. DIESEL PUMP EXHAUST LINE.

18. VACUUM TRUNK CONNECTION.

19. 10MM WEEP HOLE TO BE PROVIDED AT LOWEST POINT OF PSV OUTLET / VENT LINE.

HOLDS:

1. REQUIRED PUMP(SIZING) CONTROLS AND ALL LINE SIZING PENDING CONFIRMATION FROM THIRD PARTY FIRE CONSULTANT.

2. PRESSURE RELIEF REQUIREMENT FOR DIESEL PUMP PENDING DETAILS FROM PUMP VENDOR.

3. THIRD PARTY FIRE CONSULTANT TO CONFIRM REQUIREMENT FOR DIESEL DRIVEN PUMP.

4. DRIVER RATED POWER TO BE CONFIRMED DURING EPC.

5. REQUIREMENT FOR FIRE BRIGADE CONNECTION POINT.

6. VENT LINE SIZE TO BE CONFIRMED BY DIESEL PUMP VENDOR.

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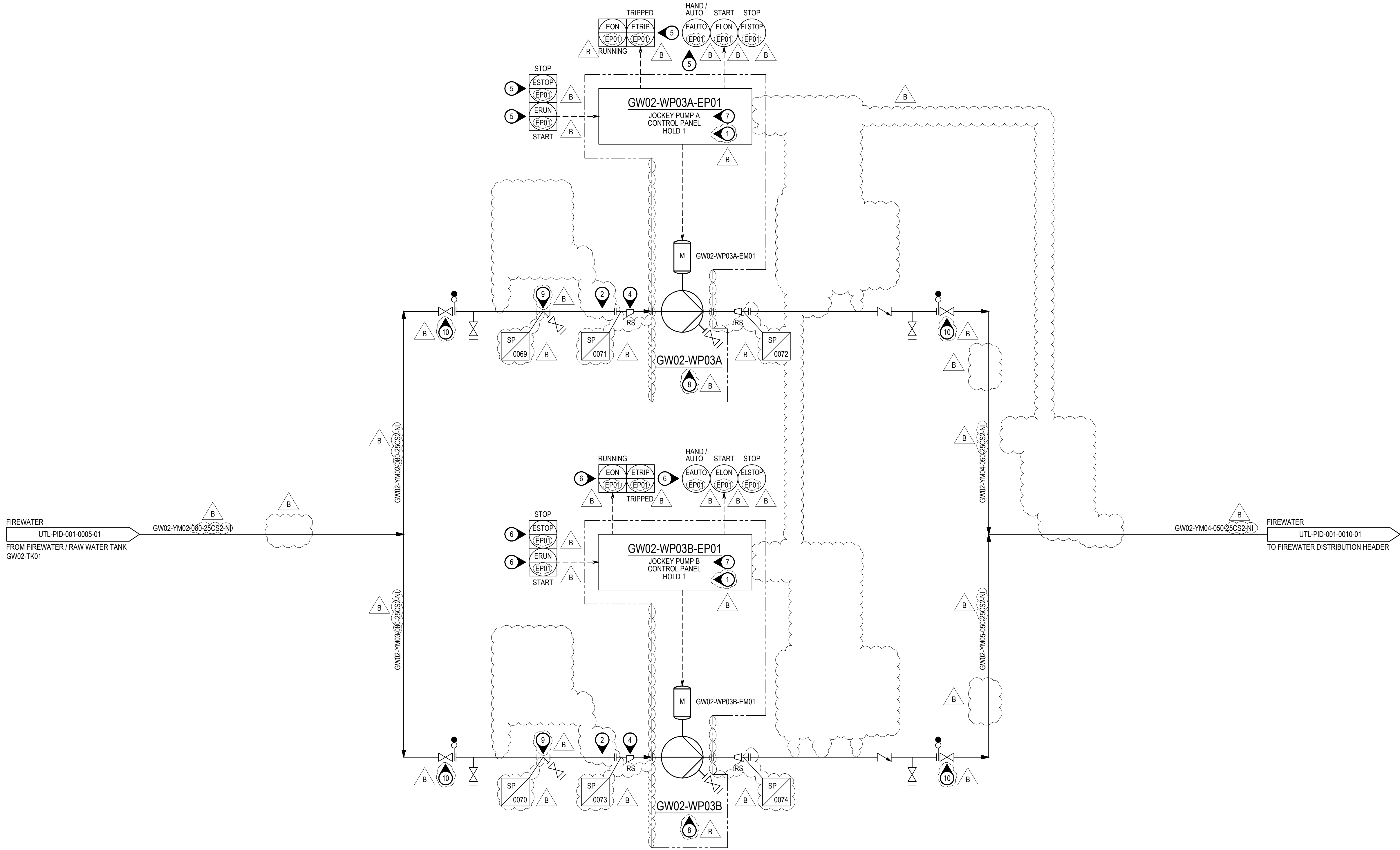
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REV	DATE	STATUS	REVISION DESCRIPTION	BY	CHK	APP	CLIENT

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PIPING AND INSTRUMENTATION DIAGRAM
FIREWATER PUMPS

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-UTL-225-PID-001-0010-01	A1	B

FILE NAME: G3ZR-UTL-225-PID-001-0010-01_B.dwg



GW02-WP03A/B 3
FIREWATER JOCKEY PUMP A/B (HOLD 1) B
DESIGN CAPACITY: 6 m³/h
RATED DIFFERENTIAL HEAD: 121.4 m
DRIVER RATED POWER: 4 kW (HOLD 3)
IMPELLER MATERIALS: 12% Cr
CASING MATERIALS: CS
INSULATION: (NONE)
TRIM: YM17-10CS2(NI)(GW02-WP03 A)
YM18-10CS2(NI)(GW02-WP03 B)

GENERAL NOTES:

1. FOR SYMBOLS, ABBREVIATIONS AND GENERAL NOTES, REFER TO LEGEND SHEETS
G3ZR-COM-225-PID-001-0001-01 TO G3ZR-COM-225-PID-001-0008-01.

NOTES:

1. ON LOW HEADER PRESSURE, FIREWATER JOCKEY PUMP SHALL START AUTOMATICALLY JOCKEY PUMP TO STOP WHEN HEADER PRESSURE RECOVERS.
2. <100 STRAIGHT LENGTH PIPE FROM PUMP INLET PER NFPA 20. / B
3. FIREWATER JOCKEY PUMPS ARE 2X100% (ONE DUTY AND ONE STANDBY). (DATA SHOWN IS FOR A SINGLE PUMP, HOLD 2)
4. ECCENTRIC REDUCER IN ACCORDANCE WITH NFPA 20. / B
5. TAGS PREFIXED GW02-WP03A
6. TAGS PREFIXED GW02-WP03B
7. CONTROL PANELS TO BE PROVIDED BY PUMP VENDOR.
8. PRESSURE SENSING LINE AND PRESSURE DETECTION ON SUCTION AND DISCHARGE OF THE PUMP TO BE INCLUDED AS PART OF VENDOR SCOPE OF SUPPLY.
9. STRAINER FOR PUMP START-UP ONLY.
10. ALL ISOLATION VALVES ON FIREWATER SERVICE LINES SHALL BE OS AND Y GATE VALVES PER NFPA 20 AND 22.

HOLDS:

1. REQUIRED PUMP SIZING, CONTROLS AND ALL LINE SIZING PENDING CONFIRMATION FROM THIRD PARTY/FIRE CONSULTANT.
2. DUTY/STANDBY REQUIREMENT TO BE CONFIRMED BY THIRD PARTY FIRE CONSULTANT.
3. DRIVER RATED POWER TO BE CONFIRMED DURING EPC.

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 **GEZERO PROJECT**



PIPING AND INSTRUMENTATION DIAGRAM
FIREWATER JOCKEY PUMPS

SCALE	DRAWING NUMBER	SIZE	REVISION
N/A	G3ZR-UTL-225-PID-001-0011-01	A1	B

FILE NAME: G3ZR-UTL-225-PID-001-0011-01_B.dwg